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# Land registration and conflict reduction: Evidence from the Brazilian Amazon☆

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ABSTRACT

Environmental land registries are increasingly used to support forest governance, yet their potential contribution to reducing rural land conflict remains poorly understood. Brazil's Cadastro Ambiental Rural (CAR) requires landholders to self-declare property boundaries and land use for environmental compliance but neither confers legal title nor resolves competing land claims. Before becoming mandatory under Brazil's 2012 Forest Code, the CAR was mandated in the Amazonian state of Pará beginning in 2008. We exploit this early implementation to examine whether an environmental land registry can reduce rural violence in frontier regions characterized by weak property rights and limited state capacity. Using municipality-level panel data (2001–2012) and a synthetic difference-in-differences design, we estimate that early CAR implementation reduced land conflict events by 11.5%, families affected by conflict by 33.4%, and land-related death threats by 10.0%. Murders and attempted murders are too rare for the design to detect effects of plausible magnitude. The estimated reductions are not uniform across outcomes or institutional environments. Death threats decline consistently across all estimation samples, whereas reductions in conflict events and families affected are concentrated in municipalities with complementary governance programs and stronger pre-treatment enforcement capacity. These findings are consistent with environmental registration operating through complementary informational and institutional channels. Documenting land occupation appears sufficient to discourage lower-intensity forms of conflict, while larger reductions in realized conflict require institutions capable of acting on the information the registry provides. More broadly, the results suggest that information-based land administration can improve land governance before formal property rights are resolved.

## 1. Introduction

This paper examines whether environmental land registries can reduce rural conflict. Environmental land registries are typically introduced to improve environmental monitoring, regulatory compliance, and land-use planning. However, they also change how

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information about land occupation is created, verified, and shared. By making land claims more visible to governments, financial institutions, courts, and other stakeholders, registries may discourage opportunistic contestation even when they neither confer legal title nor resolve competing ownership claims. In this way, a registry designed for environmental compliance may also function as an instrument of governance. Whether environmental land registries generate these broader governance benefits remains largely unknown.

The question is increasingly important because land governance has become central to both environmental and development policies. Where property rights are insecure, landholders face weaker incentives to invest, stronger incentives to exploit natural resources before others do, and greater incentives to rely on coercion to defend or expand territorial claims (Alston et al., 1996, 1999, 2000, 2012; Janvry et al., 2015). These institutional failures contribute to deforestation, carbon emissions, lower agricultural investment, and persistent land conflict throughout tropical frontiers (Hidalgo et al., 2010; Fetzer and Marden, 2017; Souza-Rodrigues, 2019; Lipscomb and Prabakaran, 2020; Albert et al., 2021; Assunção et al., 2023c; Araujo et al., 2025; Szerman et al., 2026).

This shift is not unique to Brazil.[1] Across tropical countries, governments are increasingly investing in geospatial land and environmental registries to strengthen forest governance, climate policy, and land administration, even where ownership rights remain contested or unresolved. Modern climate policy increasingly depends on these systems: carbon markets, forest conservation programs, environmental enforcement, and zero-deforestation supply-chain commitments all require credible information about who occupies land and how land is used. If environmental registries also reduce conflict, their benefits extend well beyond the environmental objectives for which they were originally designed.

Despite the growing importance of environmental registries, relatively little is known about their broader governance effects. A large literature studies how property-rights reforms, land titling, environmental enforcement, and conservation incentives affect investment, land use, and deforestation (Baragwanath and Bayi, 2020; Albert et al., 2021; Moffette et al., 2021; Assunção et al., 2023c; Vieira and Dahis, 2024; Assunção et al., 2023b; Araujo et al., 2025; Baragwanath et al., 2023; Mensah et al., 2024). Much less is known about whether a registry that documents land occupation—without granting ownership—can itself reduce conflict. This distinction is important. Formal titling changes legal rights. Environmental registration primarily changes information. Whether information alone can alter conflict is ultimately an empirical question.

Brazil provides an ideal setting to study this question because its Rural Environmental Registry (*Cadastro Ambiental Rural*, CAR) is a nationwide, mandatory environmental registry that records land occupation without adjudicating ownership.[2] Created under the 2012 Forest Code, the CAR requires rural landholders to self-declare property boundaries, land use, and legally protected forest areas. Although registration does not establish ownership, it creates a standardized and publicly accessible record of land occupation that can be used by environmental agencies, financial institutions, prosecutors, and other public actors. By reducing informational frictions surrounding land occupation, the CAR may discourage opportunistic encroachment even before formal property rights are resolved.

The staggered implementation of the CAR provides temporal and geographic variation for identification. Pará established a mandatory state-level Cadastro Ambiental Rural (CAR) in 2008, four years before registration became mandatory nationwide under Brazil's 2012 Forest Code. Most other states in the Brazilian Legal Amazon implemented the registry only after the federal mandate. Mato Grosso, by contrast, operated an earlier property registry integrated with its environmental licensing system rather than a standalone CAR. This staggered rollout allows us to compare municipalities exposed to mandatory environmental registration with otherwise similar municipalities that adopted the registry later.

We construct a municipality-level panel covering 2001–2012 by combining land-conflict records from the Pastoral Land Commission (*Comissão Pastoral da Terra*, CPT), registration data from the CAR, and data on environmental enforcement, climate, land use, and socioeconomic characteristics. We estimate treatment effects using synthetic difference-in-differences, comparing municipalities in Pará with municipalities in the remaining Legal Amazon, excluding Mato Grosso. Because CAR implementation coincided with broader environmental policies, we distinguish the effects of registration from intensified federal enforcement under the Priority Municipalities Program (*Municípios Prioritários*, MP)[3] and from the Green Municipalities Program (*Programa Municípios Verdes*, PMV), which strengthened municipal environmental governance and expanded CAR implementation.

Our estimates suggest that municipalities exposed to CAR registration experienced an 11.5% decline in land conflict events, a 33.4% reduction in families affected, and a 10.0% decline in land-related threats. Murders and attempted murders are sufficiently rare that our design is underpowered to detect effects of plausible magnitude. The reductions are strongest for lower-intensity forms of conflict. Death threats decline even after excluding municipalities affected by concurrent federal enforcement and remain significant where neither federal blacklisting nor Green Municipality support operated, suggesting that greater transparency alone discouraged opportunistic contestation. By contrast, larger reductions in realized conflicts occur primarily where complementary institutions strengthened implementation and enforcement.

[1] Comparable registry-based approaches have been adopted in several tropical countries, although they differ substantially in their institutional design and legal effects. Indonesia's One Map initiative integrates geospatial information on land claims and forest concessions to reduce overlapping claims and improve forest governance, while Colombia's Multipurpose Cadastre combines parcel mapping with land administration and restitution efforts. Peru, Mexico, and several Sub-Saharan African countries have also expanded geospatial land registration to improve tenure security and environmental governance.

[2] The CAR is a mandatory electronic registry administered by state environmental agencies and coordinated through the national SICAR system. Registration is self-declaratory and required for compliance with the Forest Code, but it neither establishes legal ownership nor adjudicates competing property claims.

[3] Beginning in 2008 the federal government identified municipalities responsible for the highest levels of Amazon deforestation as estimated by official PRODES data. These municipalities faced intensified environmental monitoring, credit restrictions, and enforcement by the Brazilian Institute of Environment and Renewable Natural Resources (*Instituto Brasileiro do Meio Ambiente e dos Recursos Naturais Renováveis*, IBAMA).

The paper then examines why environmental registration may reduce conflict. We evaluate two complementary mechanisms. The first operates through information. By making land claims more visible and easier to verify, the CAR reduces uncertainty over occupation and lowers the expected returns to contesting land through coercion. The second operates through institutional complementarity. Information alone may discourage low-intensity conflict, but its effects may be amplified where local governments possess the administrative capacity to validate registration, coordinate enforcement, and act upon the information the registry provides. Our results are consistent with both mechanisms, suggesting that environmental registries and institutional capacity function as complements rather than substitutes.

This paper contributes to several literatures. First, it shows that environmental land registries can reduce conflict even without conferring legal title, expanding the set of institutional tools available where formal land titling is politically or administratively constrained (Oliveira and Puppim, 2008; Hidalgo et al., 2010; Albertus et al., 2018; Lipscomb and Prabakaran, 2020; Mueller, 2022). Second, it contributes to research on environmental governance by showing that a policy designed to improve regulatory compliance may generate governance benefits beyond its intended environmental objective (Ali et al., 2014; Bragança and Dahis, 2022; Vieira and Dahis, 2024; Cisneros and Kis-Katos, 2024; Araujo et al., 2025). Third, it contributes to the literature on informational governance by showing that making land claims more visible changes behavior even without formal adjudication, while complementary institutions amplify these effects (Arnot et al., 2011; Ali et al., 2014; Janvry et al., 2015; Santiago et al., 2018; Jayachandran et al., 2017; Moffette et al., 2024). More broadly, our findings suggest that environmental registries can generate governance benefits alongside environmental ones, an implication that becomes increasingly important as governments rely on registry-based approaches to forest conservation, carbon markets, and climate policy.

The remainder of the paper is organized as follows. Section 2 describes the institutional background and the implementation of the CAR. Section 3 presents the conceptual framework. Section 4 describes the data and variable construction. Section 5 outlines the empirical strategy. Section 6 presents the main results, Section 7 examines the mechanisms, and Section 8 concludes.

## 2. Background

Brazil's Rural Environmental Registry (*Cadastro Ambiental Rural*, CAR) emerged as part of a broader transformation in environmental governance across the Amazon. Designed to support implementation of the Forest Code, the registry created a standardized geospatial record of rural land occupation that could be used for environmental monitoring, regulatory compliance, and land administration. Pará occupies a unique position within this transformation because it became the first Brazilian state to require CAR registration in 2008 under Decree No. 1,148/2008, four years before the registry was mandated nationally. During the same period, Pará was also a principal target of the federal Priority Municipalities Program (*Municípios sob Ação Prioritária*, MP), which intensified environmental enforcement in high-deforestation municipalities, and subsequently launched the Green Municipalities Program (*Programa Municípios Verdes*, PMV) to expand CAR implementation through technical assistance and municipal coordination. This sequence of policies provides the institutional setting for our empirical analysis. Throughout the paper, we use MP and PMV to distinguish the informational role of registration from complementary enforcement and governance interventions. Appendix H documents the spatial rollout of the CAR (Fig. H.9a), MP (Fig. H.9b), and PMV (Fig. H.9c).

The CAR is a geospatial registry that supports compliance with Brazil's Forest Code by requiring rural landholders to self-declare and georeference property boundaries, land use, and legally protected forest areas. Unlike conventional land-titling programs, the CAR records land occupation rather than adjudicating ownership. This distinction is central to our analysis because it isolates the governance value of information from the legal effects of formal property rights. The Forest Code, originally enacted in 1934 and substantially revised in 1965, established conservation requirements including Permanent Preservation Areas and Legal Reserves, which may cover up to 80% of private holdings within the Legal Amazon. For decades, however, implementation was constrained by the absence of a unified registry capable of linking environmental obligations to individual properties. The 2012 Forest Code (Law No. 12,651) addressed this institutional gap by making CAR registration mandatory nationwide and creating mechanisms for environmental regularization (Soares-Filho et al., 2014; The Nature Conservancy, 2015; Nunes et al., 2024; Chiavari et al., 2023).

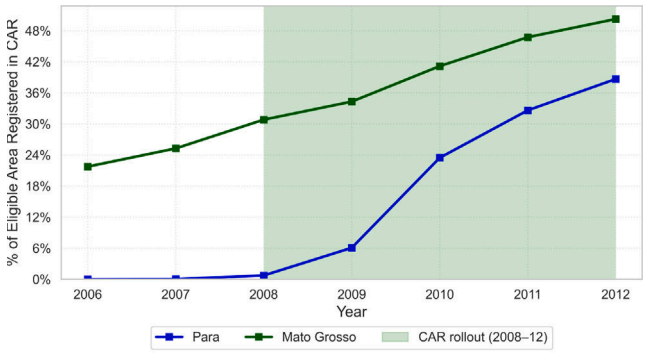
Although nationally implemented in 2012,[4] the CAR originated through earlier state initiatives. Pará piloted the registry in 2006 and made registration mandatory in 2008,[5] whereas Mato Grosso developed an earlier registry primarily for environmental licensing purposes and only later transitioned toward mandatory registration.[6] These distinct institutional trajectories matter because they generated substantial differences in the timing, coverage, and administration of environmental registration across states. For this reason, Mato Grosso is excluded from our empirical comparison, while the remaining Legal Amazon states provide the control group for our analysis.

Implementation differed markedly across the two states. As Fig. 1 shows, CAR coverage in Pará increased from less than 1% of eligible land in 2006 to nearly 39% by 2012. During the initial years of implementation, registration was concentrated among larger rural properties, with the average registered property exceeding 2300 hectares in 2010 before declining to below 600 hectares by 2012 as progressively smaller landholders entered the registry. Mato Grosso followed a considerably more gradual trajectory, reflecting its earlier integration of environmental registration into licensing rather than a dedicated registry program. By late 2023, approximately 71.8% of registrable land in Pará had validated CAR records compared with about 38% in Mato Grosso (Lopes et al., 2023, 2024).

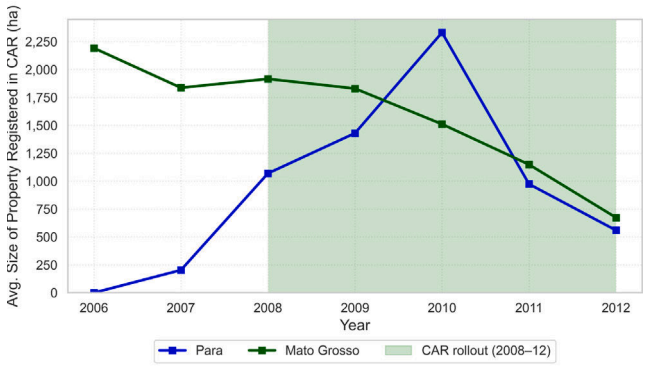
[4] Federal Law No. 12,651/2012 and Decree No. 7,830/2012 established the national CAR.

[5] Pará made parcel registration under the CAR mandatory under Decree No. 1148/2008.

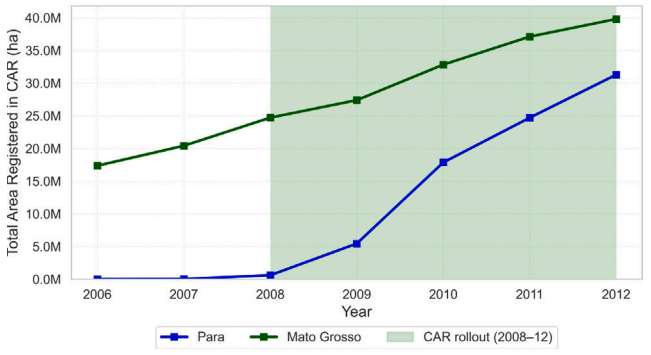
[6] Mato Grosso's SLAPR expansion in 2009, implemented under the state's environmental framework (Complementary Law No. 38/1995), served licensing purposes.




**(a) Share of eligible land registered under the CAR. Registration expanded rapidly in Pará following the 2008 mandate.**




**(b) Average registered property size. Early registration was concentrated among larger holdings before expanding to progressively smaller landholders.**




**(c) Total area registered in Pará and Mato Grosso. Registration expanded more rapidly in Pará following mandatory implementation.**

**Fig. 1.** Implementation of the rural environmental registry (CAR) in Pará and Mato Grosso. Together, the panels illustrate the rapid expansion of registration in Pará following the 2008 mandate and the distinct implementation trajectory relative to Mato Grosso.

These differences reflect not only policy timing but also substantial differences in implementation capacity, including interagency coordination, integration with land-regularization programs, and sustained institutional support.

Pará's rapid expansion of the CAR coincided with a broader transformation in Amazon environmental governance. Beginning in 2004, the federal Action Plan for the Prevention and Control of Deforestation in the Legal Amazon (Plano de Ação para Prevenção e

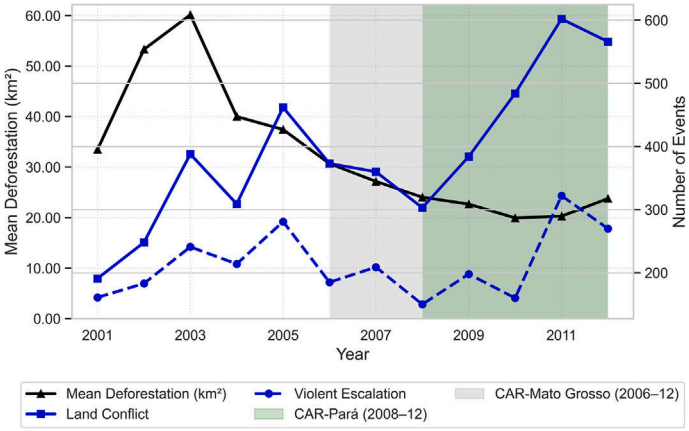



**Fig. 2.** Mean deforestation and land conflict in the Legal Amazon. Deforestation declined following the introduction of PPCDAm, whereas land conflict followed a different trajectory, motivating the question of whether environmental registration generated governance benefits beyond reductions in deforestation.

Controle do Desmatamento na Amazônia Legal, PPCDAm) combined satellite monitoring, command-and-control enforcement, credit restrictions, and institutional coordination to reduce illegal deforestation across the region (Assunção et al., 2015; Cisneros et al., 2015; Assunção and Rocha, 2019; Assunção et al., 2023a; Baragwanath and Shinde, 2025; Shinde et al., 2025). One of its principal instruments was the Priority Municipalities Program (*Municípios sob Ação Prioritária*, MP), introduced in 2008, which identified municipalities with persistently high deforestation for intensified monitoring, enforcement, and restrictions on access to subsidized rural credit. Registration in the CAR became an important component of this broader policy environment because compliance with environmental regulations increasingly depended on documenting land occupation.

Between 2008 and 2012, 20 municipalities in our estimation sample were included in the MP blacklist, 11 of which were located in Pará. These municipalities experienced substantially greater federal enforcement activity than other municipalities in the Legal Amazon. IBAMA issued, on average, approximately 17 environmental fines per municipality-year in Pará during this period, with cumulative penalties exceeding R$15.6 million (2019 BRL), while some municipalities received more than 1000 fines over the sample period. At the state level, Pará complemented these federal initiatives through the Green Municipalities Program (*Programa Municípios Verdes*, PMV), launched in 2010 to accelerate CAR implementation by providing municipalities with technical assistance, geospatial support, and institutional coordination. Together, these policies substantially increased CAR registration, particularly in municipalities facing high rates of deforestation and long-standing land disputes.

These overlapping interventions create both an opportunity and an empirical challenge. On one hand, they accelerated the expansion of environmental registration and strengthened its institutional relevance. On the other hand, they imply that any observed changes in land conflict could reflect not only registration itself but also concurrent changes in environmental enforcement or local governance. Throughout the empirical analysis, we therefore distinguish the role of CAR registration from these complementary policies by estimating specifications that exclude MP municipalities, PMV municipalities, and both programs simultaneously. These comparisons allow us to assess whether the estimated association between CAR implementation and land conflict persists outside municipalities directly exposed to these complementary interventions.

The broader evolution of environmental governance and rural conflict further motivates this distinction. As Fig. 2 illustrates, mean deforestation declined steadily after the introduction of PPCDAm, whereas land conflict followed a markedly different trajectory. Conflict events and the number of affected families increased through the late 2000s before beginning to decline only after broader CAR implementation. Although these aggregate trends are not interpreted causally, they illustrate that improvements in environmental outcomes did not automatically translate into improvements in rural governance. This divergence motivates the central question of the paper: can an environmental registry contribute to reducing land conflict by making land claims more transparent, even when formal property rights remain unresolved?

Pará's early adoption of the CAR therefore provides useful policy variation for studying the relationship between environmental registration and rural conflict. Relative to other Legal Amazon states, Pará introduced mandatory registration four years before the national rollout, while Mato Grosso followed a distinct institutional trajectory and is therefore excluded from the comparison group. Because Pará is the only state receiving treatment in 2008, other contemporaneous state-level changes cannot be ruled out completely. Rather than assuming away these factors, the empirical strategy explicitly examines their potential influence through synthetic difference-in-differences, multiple subsample analyses, extensive robustness checks, and a geographic regression discontinuity design discussed in Appendix G.1, Appendix G and Appendix E.2. The next section develops the conceptual framework linking environmental registration to land conflict through informational and institutional channels.

## 3. Conceptual framework

The conceptual framework builds on the literature linking insecure property rights, land-use decisions, and conflict in frontier economies. In these settings, weakly defined property rights increase the returns to establishing de facto control over land while

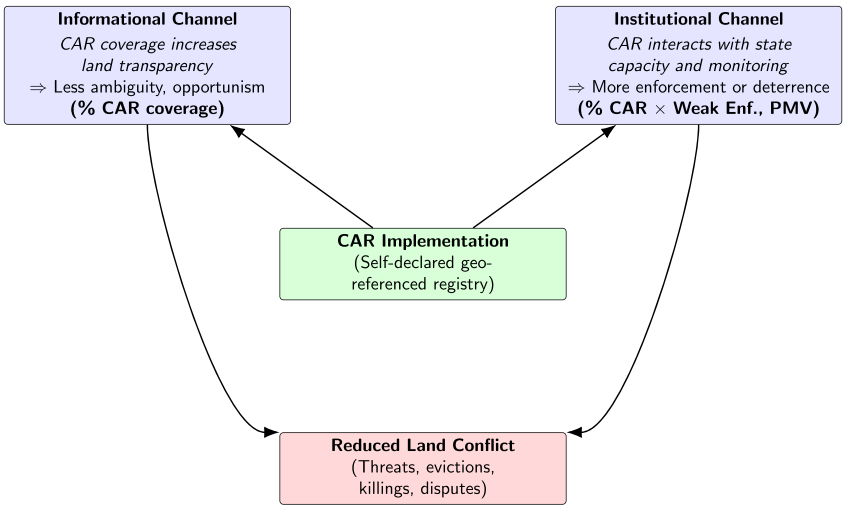



**Fig. 3.** Conceptual framework: CAR implementation may reduce land conflict through informational and institutional channels.

reducing the credibility of competing claims. Landholders therefore have incentives not only to invest and produce, but also to defend their occupation through strategic behavior that may include forest clearing, territorial expansion, or coercion (Alston et al., 1999, 2000; Amacher et al., 2009; Alston et al., 2012; Hidalgo et al., 2010; Fetzer and Marden, 2017; Schons et al., 2019; Lipscomb and Prabakaran, 2020; Baragwanath and Shinde, 2026). Environmental land registries do not alter these incentives by reallocating legal ownership. Rather, they may change the information available to governments, financial institutions, courts, and competing claimants. The central question is therefore whether improving the visibility of land occupation can change strategic behavior even before formal property rights are resolved.

Following Alston et al. (2000), land conflict arises when multiple parties contest control over the same parcel before formal adjudication. Under insecure tenure, occupation itself becomes a signal of ownership, encouraging claimants to establish possession through productive use, land clearing, or physical occupation while attempting to deter rival claimants through coercion. These incentives help explain why frontier expansion, deforestation, and land conflict frequently evolve together throughout the Amazon (Alston et al., 1999, 2000; Amacher et al., 2009; Hidalgo et al., 2010; Fetzer and Marden, 2017).

The CAR was introduced into this institutional environment. Unlike conventional land-titling programs, the registry documents occupation rather than adjudicating ownership. Registration therefore changes the information available to economic agents without directly reallocating legal property rights. This distinction is central to our analysis because it allows us to examine whether information alone can influence conflict in settings where ownership remains uncertain. We consider two complementary channels through which environmental registration may affect land conflict: an informational channel and an institutional channel. Fig. 3 summarizes these mechanisms.

### 3.1. Mechanisms

**Informational channel.** The CAR may reduce conflict by increasing the visibility and credibility of documented land occupation. Registration transforms privately held information regarding parcel boundaries and land use into a standardized administrative record that can be observed and verified by regulators, financial institutions, courts, prosecutors, and competing claimants. In this sense, the registry makes land claims more *legible*: occupation becomes publicly documented rather than known only to the occupant.

Greater transparency reduces the set of parcels over which competing claims remain sufficiently ambiguous to make opportunistic occupation attractive. As uncertainty over occupation declines, the expected return to coercive appropriation also declines because rival claimants face a greater probability that documented occupation will be recognized during administrative or legal review (Ali et al., 2014; Janvry et al., 2015; Moffette et al., 2024; Furumo et al., 2024; Mastrangelo et al., 2025).

Three complementary mechanisms may reinforce this informational effect. First, *validation*: once registered, land occupation becomes more credible to environmental agencies, courts, and other public institutions responsible for resolving disputes (Arnot et al., 2011; Ali et al., 2014). Second, *credit*: because access to rural credit increasingly depends upon CAR registration, registered landholders possess stronger incentives to maintain documented occupation while lenders can verify registration status when evaluating borrowers (Janvry et al., 2015; Assunção et al., 2020; Moffette et al., 2024). Third, *environmental monitoring*: CAR geospatial information facilitates satellite monitoring and environmental enforcement, increasing the expected costs of occupying disputed land (Alix-Garcia et al., 2018; Jung et al., 2022; Baragwanath and Shinde, 2026).

This reasoning generates the first empirical prediction:

> *Prediction 1. Municipalities with greater CAR registration should experience fewer land conflicts because registration reduces informational frictions surrounding land occupation.*

**Institutional channel.** Information alone may be insufficient to alter behavior if public institutions lack the capacity to verify, process, or act upon registered information. The effectiveness of environmental registration therefore depends not only on the existence of information but also on the institutional environment within which that information is used. Current research argues that tenure security depends as much upon expectations regarding future institutional recognition as upon current legal ownership (Deininger and Feder, 2009; Arnot et al., 2011; Janvry et al., 2015; Baragwanath and Shinde, 2026).

We examine two dimensions of this institutional environment. The first concerns enforcement capacity. Registered information is expected to have greater practical value where environmental agencies possess the resources to monitor compliance and act upon documented claims. The second concerns institutional coordination. Programs such as the Green Municipalities Program (PMV) strengthened municipal capacity to implement the CAR, coordinate geospatial information, and support environmental governance. Rather than viewing information and institutions as substitutes, we consider them complementary inputs into conflict reduction.

These considerations generate a second empirical prediction:

> *Prediction 2. The association between CAR registration and conflict reduction should be stronger where institutional capacity allows registered information to be incorporated into environmental governance and dispute resolution.*

The conceptual framework also suggests that these mechanisms need not operate uniformly across conflict outcomes. Lower-intensity forms of contestation, such as threats and intimidation, respond primarily to expectations about future disputes and therefore may react more quickly to improved information. By contrast, realized conflicts involving occupations, evictions, or violence are more likely to depend upon complementary institutional capacity and enforcement. The empirical analysis therefore distinguishes between lower- and higher-intensity conflict outcomes while examining how institutional context shapes the estimated relationship between CAR implementation and rural conflict.

## 4. Data

Our analysis uses a newly assembled municipality-level panel covering the Brazilian Legal Amazon over 2001–2012. The panel combines administrative records on land conflict, environmental registration, environmental enforcement, land use, climate, and local socioeconomic conditions. The treatment group consists of the 144 municipalities in Pará, where mandatory CAR registration began in 2008, while the comparison group comprises the remaining municipalities in the Legal Amazon after excluding Mato Grosso because of its distinct and earlier registration trajectory. The resulting balanced panel contains 632 municipalities observed over 12 years, yielding 7584 municipality-year observations. Throughout the paper, all regression tables report both municipality-year observations and the corresponding number of municipalities included in each specification.

The data are organized around four components of the empirical analysis. First, we construct municipality-level measures of land conflict that serve as the primary outcomes. Second, we define treatment using the timing of Pará's mandatory CAR implementation rather than individual registration dates. Third, we assemble variables used to examine the informational and institutional channels through which registration may influence conflict. Finally, we compile a parsimonious set of time-varying covariates that account for differences in economic conditions, demographics, climate, and protected land while avoiding controls that may themselves respond to CAR implementation.

### 4.1. Land conflict outcomes

Our primary outcome variables measure five dimensions of rural land conflict using annual reports published by the Pastoral Land Commission (*Comissão Pastoral da Terra*, CPT). The CPT documents conflicts involving the possession, use, and ownership of rural land, including disputes involving settlers, squatters, Indigenous communities, quilombolas, smallholders, landless workers, and forest-dependent populations. Because the CPT records both the occurrence and intensity of conflict, it provides unusually rich information on rural violence across the Brazilian Amazon.

We construct municipality-year measures of (i) land conflict events, (ii) families affected by conflict, (iii) death threats, (iv) attempted murders, and (v) murders. Together these outcomes span a continuum from lower-intensity intimidation to severe physical violence, allowing us to examine whether environmental registration affects different forms of conflict differently.

The original CPT reports were digitized from scanned annual publications using Python-based optical character recognition.[7] All records were manually verified against published annual reports.

Our baseline specifications estimate treatment effects using synthetic difference-in-differences with the $\log(1 + y)$ transformation, which retains municipalities with zero conflicts while reducing the influence of extreme observations in highly skewed count data. To examine the robustness of the results to alternative outcome definitions and estimation approaches, Table 2 also reports linear probability models for the extensive margin ($\mathbb{1}\{y > 0\}$), while Appendix E presents additional results using raw counts and inverse hyperbolic sine transformations.

[7] We used LayoutParser (Shen et al., 2021) to extract tabular information from scanned reports before manually verifying municipality names and annual totals against published CPT summaries.

### 4.2. Treatment definition and CAR registration

Treatment is defined by the institutional timing of mandatory CAR implementation. The treatment indicator equals one for municipalities located in Pará from 2008 onward and zero otherwise. Defining treatment at the municipality-year level reflects the policy itself rather than the timing of individual property registrations, which are incompletely recorded in the national SICAR database and therefore unsuitable for constructing consistent treatment dates.

Although treatment is defined at the state level, we also assemble annual information on CAR implementation from Jung et al. (2017); Alix-Garcia et al. (2018); Jung et al. (2022). These data include total registered area, the share of eligible land registered (% CAR coverage), and average registered property size for Pará and Mato Grosso between 2006 and 2012. These variables are used to characterize implementation and to examine the informational mechanism through which registration may influence conflict rather than to define treatment itself.

Fig. 1(b) illustrates that early registration was concentrated among larger rural properties before expanding to progressively smaller landholders after 2010. Rather than treating this pattern as a source of endogeneity, we interpret it as part of the institutional rollout of the CAR, reflecting the stronger incentives faced by larger commercial producers to comply with emerging environmental regulations.

### 4.3. Mechanism variables

The empirical analysis examines two mechanisms proposed in Section 3. The informational channel is measured using the share of eligible land registered under the CAR (% CAR coverage), which captures the extent to which land occupation has been documented within each municipality.

Institutional conditions are characterized in two ways. First, we distinguish municipalities with relatively weak federal environmental enforcement using cumulative pre-2008 IBAMA fine incidence. Constructing this measure entirely from the pre-treatment period avoids conditioning on enforcement outcomes that may themselves respond to CAR implementation. Second, we identify municipalities participating in Pará's Green Municipalities Program (PMV), which strengthened local administrative capacity through technical assistance and coordination of CAR implementation.

These variables are not interpreted as causal treatments. Instead, they characterize the institutional environments within which environmental registration operates and allow us to examine whether the estimated relationship between CAR implementation and conflict varies systematically across those environments.

### 4.4. Covariates and imputation

The baseline specification includes a parsimonious set of time-varying covariates chosen to account for differences in local economic conditions, demographics, climate, and land availability while avoiding variables that may themselves respond to CAR implementation. These covariates comprise GDP per capita and population density (Instituto Brasileiro de Geografia e Estatística (IBGE), 2021), cumulative protected area (World Database on Protected Areas (WDPA), 2021), annual precipitation, and annual maximum temperature.[8]

We do not include IBAMA fines, rural credit, or deforestation as baseline controls because each may itself respond to CAR implementation and would therefore risk conditioning on post-treatment outcomes (Section 5.2). GDP and population data are obtained from IBGE, climate variables from TerraClimate (Abatzoglou et al., 2018), and protected-area information from the World Database on Protected Areas (IUCN Categories I–VI, harmonized to IBGE municipal boundaries).

To maximize sample coverage over the full 2001–2012 period, missing observations are imputed using a two-step procedure. We first interpolate missing values within municipalities and subsequently replace any remaining gaps with year-specific means (or overall means where necessary). Appendix K reports the imputation procedure together with before-and-after summary statistics (Table K.12).

Identification primarily comes from SDID's ability to match pre-treatment outcome trajectories rather than from conditioning on observable covariates. The covariates therefore improve precision and account for observable differences in local development, demographics, and climate without driving identification. Appendix D shows that replacing contemporaneous controls with one-year lags yields qualitatively similar results.

### 4.5. Blacklisted and green municipalities

A central empirical challenge is distinguishing the relationship between CAR implementation and rural conflict from the effects of concurrent environmental policies. To address this issue, we estimate the baseline specification across five complementary samples reported in Table 2: (i) the full estimation sample; (ii) excluding municipalities designated under the federal Priority Municipalities Program (MP); (iii) excluding municipalities participating in either MP or the Green Municipalities Program (PMV); (iv) excluding PMV municipalities only; and (v) restricting the sample to municipalities participating in PMV.

[8] These covariates are standard in studies of Amazonian land use. GDP per capita and population density capture development pressure and the opportunity cost of land, protected-area coverage proxies legal constraints on land access, and precipitation and maximum temperature capture climatic conditions affecting agricultural production and land-use feasibility (Assunção et al., 2015; Assunção and Rocha, 2019; Assunção et al., 2020, 2023a; Jung et al., 2022; Anderson et al., 2018; Rosa et al., 2024).

The MP program subjected high-deforestation municipalities to intensified environmental monitoring, credit restrictions, and federal enforcement beginning in 2008. We identify 20 blacklisted municipalities using Ministry of Environment bulletins and remove them from both treated and comparison groups where appropriate. PMV, by contrast, strengthened municipal implementation of the CAR through technical assistance, institutional coordination, and geospatial support. Comparing estimates across these five samples allows us to assess whether the estimated relationship between CAR implementation and land conflict depends upon concurrent enforcement activities or broader institutional support rather than environmental registration alone. Maps of both programs are reported in Appendix H.

### 4.6. Summary statistics

Table 1 reports summary statistics for the full sample and for municipalities participating in PMV. A more detailed four-way comparison of the full sample, MP municipalities, PMV municipalities, and municipalities participating in neither program is reported in Appendix Table A.1.

**Table 1**
Summary statistics (Full sample and PMV).

| | Panel A: Full Sample | Panel B: PMV | |
|---|---|---|---|
| | Full | Control | Treated |
| Land Conflicts (Count) | 0.503 | 0.440 | 0.871 |
| | (1.599) | (1.564) | (1.741) |
| | [7584] | [6468] | [1116] |
| Families in Conflict (Count) | 40.851 | 29.718 | 105.373 |
| | (174.675) | (132.777) | (316.806) |
| | [7584] | [6468] | [1116] |
| Attempted Murders | 0.036 | 0.020 | 0.126 |
| | (0.385) | (0.295) | (0.703) |
| | [7584] | [6468] | [1116] |
| Murders | 0.038 | 0.017 | 0.159 |
| | (0.301) | (0.166) | (0.663) |
| | [7584] | [6468] | [1116] |
| Threats | 0.226 | 0.156 | 0.630 |
| | (1.303) | (1.180) | (1.812) |
| | [7584] | [6468] | [1116] |
| GDP pc (R$2019) | 5776.905 | 5631.571 | 6620.125 |
| | (5,690.182) | (4,793.925) | (9,281.041) |
| | [6938] | [5918] | [1020] |
| Fine Count | 11.124 | 9.481 | 20.647 |
| | (32.244) | (31.175) | (36.436) |
| | [7584] | [6468] | [1116] |
| Total Fines (millions R$2019) | 5.680 | 3.078 | 20.758 |
| | (48.594) | (27.812) | (106.332) |
| | [7584] | [6468] | [1116] |
| Rainfall (mm) | 162.231 | 159.156 | 180.019 |
| | (46.037) | (46.709) | (37.269) |
| | [7572] | [6456] | [1116] |
| Max Temp (K) | 316.239 | 316.307 | 315.841 |
| | (7.312) | (7.459) | (6.383) |
| | [7572] | [6456] | [1116] |
| Cumulative protected area (ha) | 1214.385 | 989.351 | 2518.614 |
| | (5,951.633) | (4,733.109) | (10,438.926) |
| | [7584] | [6468] | [1116] |
| Pop. Density | 26.213 | 23.828 | 40.049 |
| | (133.986) | (95.284) | (263.159) |
| | [7569] | [6456] | [1113] |
| CAR Area (ha) | 79,332.386 | 23,713.000 | 109,833.340 |
| | (212,555.020) | (84,604.910) | (251,877.972) |
| | [1008] | [357] | [651] |
| CAR Eligible Area (ha) | 432,094.560 | 209,737.230 | 554,032.440 |
| | (545,161.572) | (360,128.160) | (589,367.849) |
| | [1008] | [357] | [651] |
| CAR Property Size (ha) | 938.010 | 854.066 | 984.044 |
| | (4,563.737) | (4,399.757) | (4,653.898) |
| | [1008] | [357] | [651] |
| CAR Registered (%) | 14.509 | 7.973 | 18.094 |
| | (21.073) | (12.039) | (23.921) |
| | [1008] | [357] | [651] |

Notes: Means with standard deviations in parentheses and the number of observations in square brackets. CAR Area, Eligible Area, Property Size, and Registered % are available for the state of Pará only. A four-group breakdown (Full Sample, MP, PMV, Neither MP nor PMV) is reported in Appendix Table A.1.

Three features of the data motivate the empirical strategy. First, baseline conflict is highly heterogeneous across municipalities. The full sample averages approximately 0.50 land conflict events and 41 affected families per municipality-year, whereas MP municipalities average 2.18 conflict events and 199 affected families. PMV municipalities also exhibit substantially higher baseline conflict than the overall sample, while municipalities participating in neither program experience comparatively low levels of conflict. Treated municipalities are therefore systematically more conflict-prone before CAR implementation, motivating an estimator that explicitly balances pre-treatment outcome trajectories rather than relying on unconditional parallel trends.

Second, all five outcome variables are highly right-skewed, with many municipality-years recording no conflict and a relatively small number experiencing severe violence. Standard deviations range from approximately three to eleven times the corresponding means, motivating the use of $\log(1+y)$ transformations in the baseline specification together with the count, inverse hyperbolic sine, and extensive-margin robustness checks reported in the Appendix.

Finally, municipalities participating in PMV exhibit higher GDP per capita, greater protected-area coverage, higher rainfall, and more intensive environmental enforcement than municipalities outside the program. These systematic differences underscore the importance of constructing an appropriate comparison group through SDID rather than comparing treated and untreated municipalities directly.

## 5. Empirical strategy

We estimate how early implementation of the CAR is associated with subsequent changes in rural land conflict. Identification exploits the fact that Pará required mandatory CAR registration in 2008, four years before the national rollout under Brazil's 2012 Forest Code. The panel covers municipalities in the Brazilian Legal Amazon over 2001–2012 at the municipality-year level: 144 treated municipalities in Pará and 488 comparison municipalities elsewhere in the Legal Amazon, excluding Mato Grosso because of its earlier and institutionally distinct registration system.

Estimating this relationship is challenging because municipalities in Pará differed systematically from the rest of the Legal Amazon before CAR. Prior to 2008 they experienced substantially higher levels of land conflict, deforestation, and environmental enforcement. A conventional two-way fixed effects estimator relying on unconditional parallel trends is therefore unlikely to provide a credible counterfactual. We employ the Synthetic Difference-in-Differences (SDID) estimator of Arkhangelsky et al. (2021), which combines synthetic control weighting with difference-in-differences to construct a comparison group that reproduces the pre-treatment trajectory of the treated municipalities. The SDID estimator is particularly well suited to settings such as ours, where a small number of treated units exhibit systematic differences from potential controls before treatment. Rather than imposing parallel trends directly, SDID constructs a synthetic comparison group whose pre-treatment trajectory closely matches that observed in the treated municipalities.

### 5.1. Synthetic difference-in-differences

SDID constructs a weighted comparison group that matches the pre-treatment outcome trajectory of treated municipalities while retaining the two-way fixed-effects interpretation of a difference-in-differences design. Treatment effects are obtained by solving

$$(\hat{\tau}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \arg\min_{\tau,\mu,\alpha,\beta} \sum_{i=1}^{N}\sum_{t=1}^{T} \hat{\omega}_i \hat{\lambda}_t \left(Y_{it} - \mu - \alpha_i - \beta_t - D_{it}\tau\right)^2, \tag{1}$$

where $Y_{it} = \ln(1+y_{it})$ is the log-transformed conflict outcome for municipality $i$ in year $t$, $D_{it}$ equals one for Pará municipalities from 2008 onward, $\alpha_i$ and $\beta_t$ are municipality and year fixed effects, and $\hat{\omega}_i$ and $\hat{\lambda}_t$ are unit and time weights chosen to minimize pre-treatment prediction error. Estimated effects are reported on the log scale and converted to percentage changes via $(\exp(\hat{\tau})-1)\times 100$.

Unit weights reweight control municipalities so that their pre-treatment trajectory matches that observed in Pará, while time weights emphasize the pre-treatment periods most informative for predicting post-treatment outcomes. Relative to two-way fixed effects, SDID relaxes the unconditional parallel trends assumption by constructing a synthetic comparison group that better reproduces the pre-2008 evolution of treated municipalities. Inference uses the placebo variance estimator of Clarke et al. (2024), which re-estimates the full procedure—including weight construction—for each placebo unit; all SDID models use 1000 replications with a fixed seed.

### 5.2. Covariate adjustment

The baseline specification incorporates contemporaneous time-varying covariates using the projected SDID approach of Arkhangelsky et al. (2021) and Kranz (2022): outcomes are first residualized with respect to observed covariates before the synthetic weights are constructed, so the matching procedure operates on covariate-adjusted outcomes. The five controls are log GDP per capita, precipitation, maximum temperature, cumulative protected area, and population density. We deliberately exclude IBAMA fines, rural credit, and deforestation, each of which is plausibly affected by CAR, so that conditioning on them would control for the causal pathway; results are virtually identical under the default (unprojected) specification (Appendix F) and under one-year-lagged covariates (Appendix D).

For the mechanism analyses we instead employ the default (non-projected) specification. Projection would hold the ATT fixed regardless of covariate values, whereas entering the mechanism covariates into the optimization lets the counterfactual reflect the institutional environment. In every specification the reported estimate remains the ATT of CAR adoption: the mechanism covariates

enter the covariate block and do not replace the treatment indicator, so heterogeneity is read comparatively across specifications rather than from an interaction coefficient.

### 5.3. Outcomes and robustness estimators

Because $\ln(1 + y)$ can complicate interpretation when outcomes contain many zeros (Bellemare and Wichman, 2020), we report the incidence margin alongside the intensive margin rather than relying on the log specification alone. Panel B of Table 2 reports an extensive-margin linear probability model for $\mathbf{1}\{y > 0\}$, estimated by the Wooldridge (2023) `jwdid` estimator across the same five subsamples as the SDID estimates in Panel A, so the two margins can be read sample by sample. Appendix E adds raw counts in linear and Poisson QMLE forms, the inverse hyperbolic sine transformation, and SDID applied directly to raw counts. The log SDID is retained as the headline specification only because it is the one for which the SDID identifying assumptions and inference are established; the sign and significance pattern is common to all of them. Attempted murders and murders are rare events and are reported but treated as underpowered.

### 5.4. Identification assumptions and limitations

The identifying assumption is that, absent early CAR implementation, conflict outcomes in Pará would have evolved similarly to a weighted combination of municipalities elsewhere in the Legal Amazon, after conditioning on pre-treatment outcome dynamics and observed covariates. Because treatment occurs at the state level, this assumption cannot be tested directly and may be violated if other changes unique to Pará coincided with CAR implementation.

Rather than assuming these concerns away, we evaluate them through complementary analyses. First, SDID explicitly constructs a comparison group matching Pará's pre-treatment trajectory, and event-study estimates normalized to 2008 show no differential pre-treatment trends (Appendix C.1). Second, we re-estimate the baseline after excluding municipalities exposed to the federal Priority Municipalities Program (MP), Green Municipalities Program (PMV), and both, and separately restricting to PMV municipalities only—five subsamples reported side by side in Table 2. Third, we examine robustness to alternative outcome transformations, lagged covariates, projected and default specifications, and raw count estimators. Finally, Appendix G reports a geographic regression discontinuity across the Pará border, and Appendix G.1 reports a border-band SDID that retains all 144 treated municipalities but restricts the donor pool to control municipalities within 200, 150, 100, and 50 km of the boundary, applying the same unit and time weights.

Taken individually, none of these exercises can eliminate every alternative explanation associated with a single-state policy intervention. Because Pará is the only state treated in 2008, we cannot rule out that its timing coincided with other Pará-specific changes—in state environmental governance, land-titling administration, or commodity-cycle exposure—that also bore on conflict. Collectively, the exercises allow us to assess whether the estimated relationship remains consistent across comparison groups, estimation approaches, functional forms, and institutional settings. We therefore interpret our estimates as the effect of CAR adoption *under the identifying assumptions of the SDID design*, and treat the causal reading as conditional on those assumptions.

### 5.5. Mechanism specifications

The conceptual framework proposes two complementary channels. The informational channel is examined using municipal CAR coverage; the institutional channel is examined using its interactions with weak environmental enforcement and with PMV participation, forming Panels A, B1, and B2 of Table 3.

Any partition based on IBAMA fines must be pre-determined, since contemporaneous fines are plausibly a post-treatment outcome of CAR: the registry itself raised monitoring intensity. We therefore report the enforcement interaction both on the contemporaneous annual split (Panel B1) and on cumulative *pre-2008* fines only (Panel B1′), which cannot be an outcome of CAR; the two are identical to three decimal places. These mechanism covariates are identically zero for all control municipalities and hence near-collinear with treatment, so the counterfactual barely reweights and the interaction panels function as robustness checks on the ATT rather than as a decomposition of channels—which also means the post-treatment-fine endogeneity concern cannot bias the reported ATT. To identify enforcement heterogeneity we therefore add a direct subsample split, running the main SDID separately on weak- and strong-enforcement municipalities classified by pre-2008 cumulative fines (Table 4). Comparisons across institutional environments are interpreted as descriptive rather than as evidence of the causal effect of enforcement itself.

The following section reports the baseline SDID estimates before examining robustness and the informational and institutional patterns implied by the conceptual framework.

## 6. Main results: The effects of the CAR on land conflict

Table 2 reports average treatment effects of CAR implementation on five land conflict outcomes, estimated by synthetic difference-in-differences (Arkhangelsky et al., 2021; Kranz, 2022; Clarke et al., 2024) with placebo-based standard errors based on 1000 replications. The five columns are presented to be read against one another. Column (1) is the full sample; column (2) removes the 20 federally blacklisted (MP) municipalities, isolating the CAR from concurrent federal enforcement; column (3) removes both blacklisted and Green Municipalities (PMV) participants, leaving 530 municipalities exposed to neither program; column (4) removes only Green Municipalities (PMV); and column (5) restricts to Green Municipalities (PMV). Reading across the columns therefore separates what the registry accomplishes on its own from what it accomplishes alongside enforcement pressure and coordinated governance support. Fig. 4 and appendix Fig. C.6 show the SDID-weighted outcome paths for the full sample: treated and synthetic-control municipalities track closely over 2001–2007, and the gap narrows after 2008.

**Table 2**
Effects of CAR registration on land conflict.

| | Full sample (1) | Drop MP (2) | Drop MP & PMV (3) | Drop PMV (4) | PMV only (5) |
|---|---|---|---|---|---|
| *Panel A. Synthetic difference-in-differences; dependent variable:* $\ln(1+y)$ | | | | | |
| *Conflict intensity* | | | | | |
| Conflict events | −0.122*** | −0.100*** | −0.046 | −0.047 | −0.163*** |
| | (0.030) | (0.029) | (0.041) | (0.046) | (0.034) |
| Families affected | −0.407*** | −0.375*** | −0.272* | −0.258 | −0.483*** |
| | (0.113) | (0.111) | (0.160) | (0.168) | (0.127) |
| Death threats | −0.106*** | −0.098*** | −0.087*** | −0.090*** | −0.113*** |
| | (0.021) | (0.020) | (0.030) | (0.031) | (0.025) |
| *Rare violent outcomes* | | | | | |
| Attempted murders | 0.010* | 0.007 | −0.013 | −0.012 | 0.024*** |
| | (0.006) | (0.006) | (0.010) | (0.010) | (0.008) |
| Murders | −0.019*** | −0.002 | −0.001 | −0.006 | −0.026*** |
| | (0.006) | (0.004) | (0.006) | (0.010) | (0.007) |
| *Panel B. Linear probability model; dependent variable:* $\mathbf{1}\{y>0\}$ | | | | | |
| *Conflict intensity* | | | | | |
| Conflict events | −0.076*** | −0.066*** | −0.034 | −0.036 | −0.097*** |
| | (0.024) | (0.025) | (0.036) | (0.036) | (0.029) |
| Families affected | −0.072*** | −0.063** | −0.035 | −0.037 | −0.092*** |
| | (0.024) | (0.025) | (0.037) | (0.037) | (0.029) |
| Death threats | −0.074*** | −0.071*** | −0.083*** | −0.084*** | −0.069*** |
| | (0.018) | (0.017) | (0.023) | (0.023) | (0.023) |
| *Rare violent outcomes* | | | | | |
| Attempted murders | −0.002 | −0.000 | −0.010 | −0.010 | −0.009 |
| | (0.010) | (0.009) | (0.007) | (0.007) | (0.015) |
| Murders | −0.028** | −0.005 | −0.014 | −0.016 | −0.035* |
| | (0.013) | (0.011) | (0.012) | (0.012) | (0.020) |
| Municipality and year fixed effects | ✓ | ✓ | ✓ | ✓ | ✓ |
| Time-varying controls (Panel A only) | ✓ | ✓ | ✓ | ✓ | ✓ |
| Observations | 7584 | 7344 | 6360 | 6468 | 6972 |
| Municipalities | 632 | 612 | 530 | 539 | 581 |
| Treated (Pará) | 144 | 133 | 51 | 51 | 93 |
| Control | 488 | 479 | 479 | 488 | 488 |

*Notes:* Observations are municipality–years, 2001–2012. Treated = Pará municipalities (*Cadastro Ambiental Rural* mandatory from 2008); controls = Legal Amazon excluding Mato Grosso. MP = 20 blacklisted Priority Municipalities (11 in Pará, 9 elsewhere); PMV = *Programa Municípios Verdes*, which operates only in Pará. Columns (2) and (3) therefore also remove 9 control municipalities, while columns (4) and (5) leave the 488 controls intact and partition the 144 treated municipalities into the 51 that never joined the PMV and the 93 that did. Column (3) drops both programs and is the most conservative sample. *Estimators differ across panels.* Panel A is the synthetic difference-in-differences ATT (Arkhangelsky et al., 2021) with projected covariates (Kranz, 2022; Clarke et al., 2024), in log points; full-sample estimates imply −11.5, −33.4, and −10.0 percent for conflict events, families, and threats via $(\exp(\hat{\tau})-1)\times 100$. Panel B is the Wooldridge (2023) `jwdid` linear probability model on $\mathbf{1}\{y>0\}$, in probability units. Differing in transformation, weighting, and inference, they corroborate independently. *Controls, inference, means.* Panel A controls for logged GDP per capita, precipitation, maximum temperature, cumulative protected area, and population density; Panel B omits covariates (logged versions in Table E.7). Standard errors: placebo-based, 1000 replications (A); clustered by municipality (B). Pre-treatment control means are 0.331, 28.561, 0.118, 0.016, and 0.012 events (A) and 0.182, 0.174, 0.034, 0.009, 0.009 as Pr(any) (B), in row order. The two rare violent outcomes are underpowered (footnote 9); we draw no conclusions from them. $^{***}p<0.01$, $^{**}p<0.05$, $^{*}p<0.10$.

Three patterns emerge. Death threats fall by a similar amount in every subsample. Conflict events and families affected fall substantially on average, but the reductions are concentrated where complementary programs were active. Murders and attempted murders are too rare for the design to detect effects of plausible size, and we do not interpret them.

**Death threats.** Threats show the most consistent pattern in the table. The full-sample ATT is −0.106 (SE = 0.021), a 10.0% reduction relative to the pre-treatment control mean. The estimate barely moves across subsamples—−9.4% excluding blacklisted municipalities, −8.3% excluding both programs, −8.7% excluding Green Municipalities, and −10.7% among Green Municipalities alone—and is statistically significant at the 1% level in all five. The extensive-margin estimates in Panel B track this closely: the probability that any threat occurs falls by between 6.9 and 8.4 percentage points, again is significant at the 1% level in every column (Table E.7).

We interpret this stability as evidence consistent with the CAR's informational effect. Threats are forward-looking intimidatory acts—warnings, coercive signals, territorial challenges—whose effectiveness depends on ambiguity about who holds a parcel.

Registration reduces that ambiguity at the moment of enrollment, before any agency acts on the record. This interpretation is consistent with the persistence of the estimated reduction even in the absence of the enforcement or administrative support that the other outcomes appear to require.

**Conflict events and families affected.** The two measures of conflict intensity behave similarly to one another and differently from threats. In the full sample, conflict events fall by 11.5% (−0.122, SE = 0.030) and the number of families affected falls by 33.4% (−0.407, SE = 0.113). Both attenuate as complementary programs are removed. Excluding blacklisted municipalities leaves reductions of 9.5% (−0.100, SE = 0.029) and 31.3% (−0.375, SE = 0.111); excluding both programs leaves 4.5% (−0.046, SE = 0.041), which is no longer statistically significant, and 23.8% (−0.272, SE = 0.160), which remains significant at the 10% level; excluding only Green Municipalities gives a comparable 22.8% (−0.258, SE = 0.168). Restricting to Green Municipalities produces the largest estimates in the table: −15.0% for conflict events (−0.163, SE = 0.034) and −38.3% for families affected (−0.483, SE = 0.127).

That conflict events lose significance in column (3) while families affected does not is informative. Families affected measures the human scale of a dispute, not the count of distinct episodes. Clarified boundaries can narrow the territory over which a dispute ranges—and so the number of households drawn into it—before they reduce the number of disputes that occur at all. The pattern across columns is consistent with the interpretation we develop in Section 7: information alone deters the lowest-intensity contestation, while reductions in realized conflict require the institutional capacity to act on what the registry reveals (Table 4).

The timing reinforces this. CAR coverage in Pará expanded from roughly 1% of eligible land in 2008 to 6.1% in 2009, 23.5% in 2010, and 38.7% by 2012 (Table B.2), and the sharp decline in conflict visible in Fig. 4 between 2009 and 2010 coincides with that expansion. Because the same decline appears in columns (3) and (4), which exclude Green Municipalities entirely, it cannot be an artifact of their entry in the program in 2010.

**Murders and attempted murders.** We report both rare violent outcomes for completeness but draw no conclusions from them. Murders fall by 1.9% in the full sample (−0.019, SE = 0.006) and 2.6% among Green Municipalities (−0.026, SE = 0.008), but the estimate is indistinguishable from zero once blacklisted municipalities are removed (−0.2%, −0.002, SE = 0.004). Attempted murders are positive in three columns (significant in the full and PMV-only samples) and negative but insignificant in two, with no systematic pattern.

Both outcomes are too rare to support inference at these magnitudes. Pre-treatment control means are 0.012 murders and 0.016 attempted murders per municipality-year, and the smallest effect the design can detect at conventional power is roughly 1.7%.[9] We therefore treat these outcomes as underpowered and base the paper's conclusions on threats, families affected, and conflict events. In particular, we do not read the positive attempted-murder coefficients as evidence of substitution from lethal to non-lethal violence.

### 6.1. Event study analysis

Fig. 4 presents dynamic treatment effects for all five outcomes, with event time normalized to 2008 and 95% bootstrap confidence intervals. Appendix C, Figs. C.1–C.5, report the same plots separately for each subsample.

Pre-treatment estimates are flat and statistically indistinguishable from zero for every outcome and in every sample, which is the condition that the SDID design requires.

Post-treatment paths differ across outcomes in a way that matches the cross-sectional results. Threats decline immediately after 2008 and stay down through 2012, consistent with a deterrent that operates from the moment of registration. Conflict events and families affected instead show a brief increase in 2008–2009 before declining from 2010 onward. We read the initial increase as a transitional effect: formalizing previously informal or overlapping claims can surface latent disputes before clarified boundaries begin to suppress them. Murders are negative but imprecise throughout, and attempted murders fluctuate around zero—both consistent with the low baseline incidence discussed above.

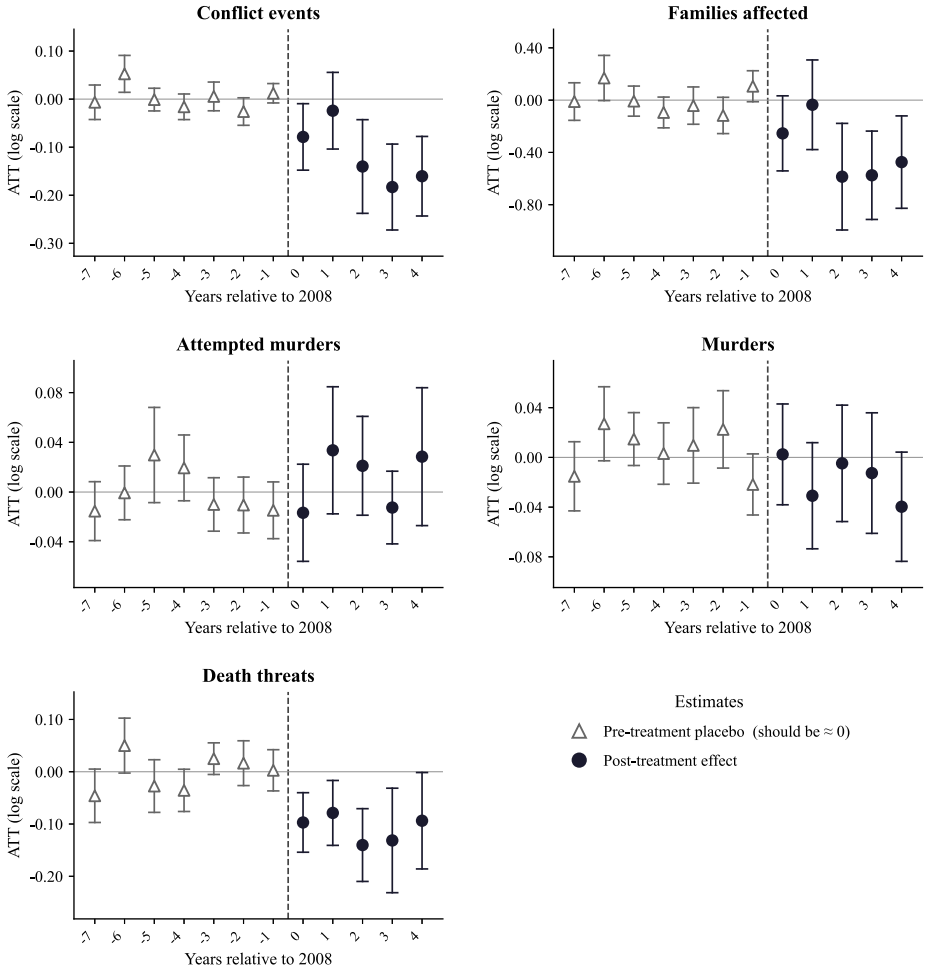
The subsample panels in Appendix C show near-identical pre-treatment dynamics and similar post-treatment paths when blacklisted municipalities are excluded, indicating that the timing of the effects is not driven by federal enforcement actions.

### 6.2. Robustness

The estimates survive a wide range of alternative specifications, reported in the appendices and summarized here. Re-estimating with one-year-lagged covariates leaves the results qualitatively unchanged (Appendix D). Modeling raw counts with the Wooldridge (2021) `jwdid` estimator under linear and Poisson QMLE forms, applying the inverse hyperbolic sine transformation, and estimating an extensive-margin linear probability model all deliver the same signs and significance pattern (Appendix E, Tables E.5 and E.7). Applying SDID directly to raw counts confirms sizeable absolute declines (Appendix E.3), and the estimates are essentially identical under the default rather than projected covariate option (Appendix F).

Two border-based designs exploit the Pará boundary directly. A geographic regression discontinuity compares municipalities on either side of the state line (Appendix G), a border-band SDID restricts the donor pool to control municipalities within a fixed distance of the border (Appendix G.1, Table G.10), thereby reproducing or strengthening the main effects.

[9] A minimum-detectable-effect calculation makes this concrete. At 80% power and a 5% two-sided significance level, the smallest log-scale effect detectable is approximately $(z_{0.975} + z_{0.80}) \times \text{SE} = 2.80\,\text{SE}$. At the reported placebo standard errors this is about 1.7% for murders (SE = 0.006) and 1.7%–2.8% for attempted murders (SE = 0.006–0.010). The estimated attempted-murder effects are below 1.3% in absolute value in four of five samples, and the murders effect (−1.9%) sits near the detectable margin and is significant only in the full and PMV-only samples.




Notes: SDID event-study ATTs, full sample. 95% bootstrap CI (1000 reps). Triangles = pre-treatment placebo (parallel trends check). Circles = post-treatment effect. Dashed = 2007.5.

**Fig. 4.** Event Study for all five outcomes using the full sample based on Table 2, column (1). Each panel shows SDID ATT estimates for one subsample. Hollow triangles = pre-treatment placebo (years −7 to −1); filled circles = post-treatment effect (years 0 to +4). Capped bars = 95% bootstrap CI (1,000 replications). Dashed line = 2007.5.

We interpret these estimates as causal only under the identifying assumptions set out in Section 5.4. Because treatment is assigned to a single state in a single year, the design cannot rule out contemporaneous Pará-specific shocks around 2008, and the border RD is a local, low-powered check whose cross-border spillovers bias it toward zero. What the appendices establish is narrower but still substantial: the findings do not depend on any single functional form, estimator, comparison group, or identification strategy.

## 7. Mechanisms

Section 3 proposed two complementary channels through which the CAR may influence rural land conflict. The informational channel operates by increasing the visibility and credibility of documented land occupation, while the institutional channel operates through the capacity of public institutions to verify and act upon that information. Section 6 established an empirical pattern that these mechanisms must explain. Death threats decline consistently across all estimation samples, including municipalities exposed to neither the federal blacklisting program nor the Green Municipalities Program (PMV). By contrast, reductions in conflict events and families affected are largest where complementary governance programs remain in place. The purpose of this section is therefore to assess whether these empirical patterns are consistent with the mechanisms proposed by the conceptual framework.

We proceed in two steps. We first examine whether conditioning the SDID estimator on the proposed mechanism variables materially changes the estimated average treatment effect. We then estimate the baseline specification separately across municipalities differing in pre-treatment enforcement capacity. The first exercise provides a diagnostic regarding identification of the proposed mechanisms, while the second offers evidence on how the estimated relationship between CAR implementation and conflict varies across institutional environments.

**Table 3**
Robustness of the ATT to conditioning on mechanism covariates.

| | Conflict intensity | | Rare violent outcomes | | |
|---|---|---|---|---|---|
| | Conflict events (1) | Families affected (2) | Attempted murders (3) | Murders (4) | Death threats (5) |
| *Panel A. Add % CAR coverage* | | | | | |
| ATT | −0.124*** | −0.426*** | 0.010* | −0.014** | −0.105*** |
| | (0.029) | (0.112) | (0.006) | (0.006) | (0.021) |
| % change | −11.7 | −34.7 | +1.1 | −1.4 | −10.0 |
| *Panel B1. Add % CAR × weak enforcement (contemporaneous)* | | | | | |
| ATT | −0.123*** | −0.409*** | 0.008 | −0.018*** | −0.104*** |
| | (0.029) | (0.112) | (0.006) | (0.006) | (0.021) |
| % change | −11.5 | −33.6 | +0.8 | −1.8 | −9.9 |
| *Panel B1′. Add % CAR × weak enforcement (pre-2008, pre-determined)* | | | | | |
| ATT | −0.123*** | −0.409*** | 0.008 | −0.018*** | −0.104*** |
| | (0.029) | (0.112) | (0.006) | (0.006) | (0.021) |
| % change | −11.5 | −33.6 | +0.8 | −1.8 | −9.9 |
| *Panel B2. Add % CAR × PMV participation* | | | | | |
| ATT | −0.124*** | −0.422*** | 0.010* | −0.015** | −0.105*** |
| | (0.029) | (0.112) | (0.006) | (0.006) | (0.021) |
| % change | −11.7 | −34.5 | +1.0 | −1.5 | −9.9 |
| Municipality and year fixed effects | ✓ | ✓ | ✓ | ✓ | ✓ |
| Base covariates (lagged) | ✓ | ✓ | ✓ | ✓ | ✓ |
| Observations | 7584 | 7584 | 7584 | 7584 | 7584 |
| Municipalities | 632 | 632 | 632 | 632 | 632 |
| Treated municipalities | 144 | 144 | 144 | 144 | 144 |

*Notes:* Each panel reports the SDID ATT on log outcomes (Arkhangelsky et al., 2021), default (non-projected) method, when the named mechanism variable is added to the five logged base covariates: GDP per capita, precipitation, temperature, cumulative protected area, and population density. Weak enforcement = below-median cumulative IBAMA fine counts. Panel B1 uses contemporaneous fines; Panel B1′ uses cumulative 2001–2007 fines, a pre-determined measure that cannot be an outcome of the CAR. *Interpretation.* The mechanism variables are zero for all control municipalities and for treated municipalities before 2008, so each is near-collinear with the treatment indicator and the synthetic counterfactual reweights only marginally. The ATT is consequently near-invariant across panels—identical to the base-covariate estimate in Panels B1 and B1′, where the fine interaction is fully absorbed. Panels B1 and B1′ are in turn numerically identical to *each other* in every column: the contemporaneous and the pre-2008 fine interactions are non-zero on exactly the same observations (treated municipalities in post-2008 years) and zero everywhere else, so both collapse to the same absorbed regressor and return the same ATT and standard error to all reported digits. This coincidence is informative rather than incidental—it shows the estimate does not depend on whether enforcement is measured contemporaneously or by a pre-determined, non-endogenous proxy. We read this table as establishing that the ATT is robust to conditioning on these covariates—not as identifying mechanism heterogeneity, for which we split the sample in Table 4. Standard errors in parentheses are placebo-based with 1000 replications (seed 2025). % change is $(\exp(\hat{\tau}) - 1) \times 100$. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

### 7.1. Mechanism covariates and identification

Table 3 reports the estimated ATT after sequentially adding the proposed mechanism variables to the baseline SDID specification. These include municipal CAR coverage, its interaction with weak environmental enforcement, and its interaction with PMV participation.

The estimated treatment effects change very little across specifications. Conflict events remain close to an 11–12% reduction, families affected at approximately 34%, and death threats at approximately 10%, regardless of the mechanism variable included. This stability indicates that conditioning on these variables does not materially alter the synthetic counterfactual or the estimated ATT.

The reason is straightforward. The mechanism variables are identically zero for every control municipality and for all treated municipalities before 2008. Consequently, they provide relatively little additional variation beyond the treatment indicator itself and therefore contribute only marginally to the construction of the synthetic comparison group. The resulting estimates should therefore be interpreted as demonstrating the robustness of the baseline ATT rather than identifying heterogeneity in the mechanisms through which CAR operates.

This finding is nevertheless informative. Contemporaneous IBAMA fines may themselves respond to CAR implementation because environmental registration increases the visibility of land use and facilitates environmental monitoring. Conditioning directly on such variables could therefore raise concerns about post-treatment bias. Table 3 indicates that this concern has little practical consequence

**Table 4**
Effects by pre-2008 enforcement Intensity.

| | Conflict intensity | | Rare violent outcomes | | |
|---|---|---|---|---|---|
| | Conflict events (1) | Families affected (2) | Attempted murders (3) | Murders (4) | Death threats (5) |
| *Panel A. Weak enforcement* (48 treated, 263 control) | | | | | |
| ATT | −0.009 | 0.010 | 0.012* | −0.002 | −0.050* |
| | (0.039) | (0.155) | (0.006) | (0.006) | (0.027) |
| % change | −0.9 | +1.0 | +1.2 | −0.2 | −4.9 |
| Observations | 3732 | 3732 | 3732 | 3732 | 3732 |
| Municipalities | 311 | 311 | 311 | 311 | 311 |
| *Panel B. Strong enforcement* (96 treated, 225 control) | | | | | |
| ATT | −0.201*** | −0.658*** | 0.011 | −0.025** | −0.137*** |
| | (0.049) | (0.174) | (0.010) | (0.012) | (0.035) |
| % change | −18.2 | −48.2 | +1.1 | −2.5 | −12.8 |
| Observations | 3852 | 3852 | 3852 | 3852 | 3852 |
| Municipalities | 321 | 321 | 321 | 321 | 321 |
| Municipality and year fixed effects | ✓ | ✓ | ✓ | ✓ | ✓ |
| Base covariates (logged, projected) | ✓ | ✓ | ✓ | ✓ | ✓ |

*Notes:* Each panel reports the ATT from the main SDID specification— projected lagged covariates, municipality and year fixed effects, log outcomes— estimated on a subsample. Municipalities are classified once, using pre-treatment information only, as weak- or strong-enforcement according to whether cumulative IBAMA fine counts over 2001–2007 fall below or above the cross-municipal median. The classification is fixed and cannot be an outcome of the CAR, which was piloted from 2008. Weak-enforcement municipalities begin from lower baseline conflict (pre-treatment control mean 0.11 against 0.25 in logs), so part of the difference in estimated magnitudes reflects less scope for measured reduction rather than a difference in response. Both subsamples retain more control than treated municipalities, so placebo inference (1,000 replications, seed 2025) applies as in the main table. % change is $(\exp(\hat{\tau}) - 1) \times 100$. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

for the estimated ATT. Replacing contemporaneous enforcement with cumulative pre-2008 IBAMA fines, which cannot themselves be affected by CAR, produces numerically identical estimates. The baseline relationship between CAR implementation and conflict is therefore not sensitive to whether enforcement is measured contemporaneously or using predetermined information.

The limitation of this exercise is equally important. Because the mechanism covariates contribute little independent variation beyond treatment itself, they cannot identify whether CAR operates differently across institutional environments. To address that question, we next estimate the baseline specification separately for municipalities differing in pre-treatment enforcement capacity.

### *7.2. Institutional complementarity*

Table 4 reports separate SDID estimates for municipalities with relatively weak and relatively strong pre-treatment environmental enforcement, classified using cumulative IBAMA fines over 2001–2007. Because the classification relies exclusively on pre-treatment information, it cannot itself be influenced by CAR implementation. Fig. H.9(b) illustrates the spatial distribution of the two enforcement groups.

The contrast across institutional environments is pronounced. Municipalities with stronger enforcement capacity experience substantially larger reductions in conflict events, families affected, and death threats than municipalities with weaker enforcement. Estimated reductions in conflict events and families affected are concentrated almost entirely among municipalities where public institutions possessed greater capacity to monitor compliance and act upon documented land claims.

Death threats present an important exception. Although the estimated reduction is smaller under weak enforcement, it remains negative and statistically significant. This pattern closely mirrors the results reported in Table 2, where reductions in death threats persist even after excluding municipalities participating in both MP and PMV. Taken together, these findings suggest that lower-intensity forms of conflict respond more directly to improvements in information, whereas reductions in realized conflict depend more heavily upon complementary institutional capacity.

An alternative explanation is that municipalities with weaker enforcement begin from lower baseline levels of conflict, reducing the scope for measurable declines. While this consideration may contribute to the smaller estimated effects for conflict events and families affected, it does not explain the persistence of reductions in death threats. The evidence is therefore most consistent with two complementary channels rather than a single mechanism operating uniformly across all conflict outcomes.

This interpretation follows naturally from the conceptual framework. Threats represent relatively low-cost, forward-looking attempts to influence expectations regarding future disputes. Once competing claims become publicly documented, the expected return to intimidation may decline immediately, even before formal enforcement occurs. By contrast, realized conflicts involving occupations, evictions, or household displacement require administrative or legal intervention to resolve and therefore depend more heavily upon institutional capacity.

### 7.3. Summary

The mechanism analysis refines rather than overturns the conceptual framework developed in Section 3. Three conclusions emerge.

First, conditioning the SDID estimator on the proposed mechanism covariates leaves the estimated ATT essentially unchanged, indicating that the baseline estimates are robust to these alternative specifications while also illustrating that this approach cannot directly identify mechanism heterogeneity.

Second, comparisons across municipalities differing in pre-treatment enforcement capacity reveal substantially larger reductions in conflict events and families affected where institutional capacity is stronger. These results are consistent with environmental registration and institutional capacity functioning as complements rather than substitutes.

Third, reductions in death threats remain remarkably stable across institutional environments and across the alternative samples considered in Section 6. This consistency suggests that improved information regarding land occupation may discourage lower-intensity forms of contestation even where complementary institutional support is limited.

Taken together, the evidence is most consistent with environmental registration affecting conflict through two complementary channels. Information appears sufficient to reduce uncertainty surrounding land occupation and discourage intimidation, while larger reductions in realized conflict emerge primarily where stronger institutions enable documented information to be translated into monitoring, dispute resolution, and environmental governance. Because the empirical design compares treatment effects across institutional environments rather than identifying formal mediation effects, these findings should be interpreted as evidence consistent with the proposed mechanisms rather than as a causal decomposition of the channels through which CAR operates.

## 8. Policy implications and conclusion

This paper examines whether an environmental land registry can improve rural governance even before formal property rights are resolved. Exploiting Pará's early implementation of the Rural Environmental Registry (CAR) and a synthetic difference-in-differences design, we estimate that municipalities exposed to early CAR implementation experienced fewer land conflict events, fewer families affected by conflict, and fewer land-related death threats. These reductions are not uniform across outcomes or institutional environments. Death threats decline consistently across all estimation samples, whereas larger reductions in realized conflict emerge primarily where environmental registration was accompanied by stronger institutional capacity and coordinated local governance.

Taken together, the evidence is most consistent with a framework in which information and institutions operate as complements rather than substitutes. Documenting land occupation appears sufficient to reduce uncertainty surrounding land claims and discourage lower-intensity forms of contestation, even where enforcement capacity is limited. However, translating improved information into reductions in realized conflict requires public institutions capable of verifying claims, coordinating administrative responses, and enforcing environmental regulations. Environmental registration therefore generates its greatest governance benefits when combined with effective local institutions rather than viewed as a substitute for them.

These findings contribute to a broader understanding of land governance in frontier regions. Much of the literature emphasizes formal land titling as the principal mechanism through which tenure security improves investment and reduces conflict (Oliveira and Puppim, 2008; Hidalgo et al., 2010; Albertus et al., 2018; Lipscomb and Prabakaran, 2020). Our results suggest that meaningful governance improvements may arise earlier in the institutional process. By making land occupation more transparent to governments, financial institutions, courts, and competing claimants, environmental registries can reduce informational frictions even when competing ownership claims remain unresolved.

The results also point to an underappreciated co-benefit of environmental policy. Although the CAR was introduced to improve compliance with Brazil's Forest Code, the registry appears to have generated broader governance benefits alongside its environmental objectives. As governments increasingly rely on geospatial registries to implement forest conservation policies, carbon markets, zero-deforestation supply chains, and other climate initiatives, the administrative information generated by these systems may also strengthen the institutional foundations necessary for more stable land governance.

Several limitations qualify these conclusions. First, the analysis evaluates a single state's early implementation of the CAR during a period of unusually strong political commitment to Amazon conservation, and the estimated relationships may not generalize to later implementation periods or institutional settings with weaker administrative capacity (Nunes et al., 2024; Furumo et al., 2024). Second, the municipality-level analysis cannot identify heterogeneous effects across different types of landholders or disputes. Third, although the empirical strategy incorporates multiple comparison groups and extensive robustness analyses, the interpretation remains conditional on the identifying assumptions discussed in Section 5.4. Finally, CPT records may underreport conflict in remote municipalities, implying that the estimated relationships are likely conservative.

More broadly, the findings suggest that governments need not wait for complete resolution of property rights before realizing meaningful improvements in land governance. Information-based land administration can itself generate governance benefits by reducing uncertainty surrounding land occupation, while stronger institutions determine how fully those informational gains translate into reductions in conflict. In frontier regions, where environmental governance and land governance are closely intertwined, investments in transparent land information systems may therefore contribute simultaneously to environmental protection, institutional development, and social stability.

## CRediT authorship contribution statement

**Nilesh Shinde:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Stella**

**Z. Schons:** Writing – review & editing,Supervision, Resources, Methodology, Funding acquisition, Conceptualization. **Alexandre G. Maia:** Supervision, Methodology, Conceptualization. **Gregory S. Amacher:** Supervision, Resources.

## Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Grammarly and ChatGPT to edit the grammar of the writing. After using these tools, the authors reviewed and edited the content as needed and took full responsibility for the content of the publication.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix A. Detailed summary statistics

Table A.1 reports summary statistics for the full sample and separately for MP (*Municípios Prioritários*), PMV (*Programa Municípios Verdes*), and municipalities in neither program, over 2001–2012, further split into treated and control observations for MP and PMV.

Baseline conflict differs sharply across groups. MP-treated municipalities are by far the most conflict-prone: they average 2.175 conflict events and 199 affected families per year, against 0.448 and 36 in MP controls and 0.418 and 28 in municipalities outside both programs. Violent outcomes follow the same pattern—murders average 0.533 in MP-treated municipalities, more than twenty times the MP-control mean (0.022), and threats average 1.608 versus 0.181. PMV-treated municipalities are also more conflict-prone than their controls (0.871 vs. 0.440 conflicts; 105 vs. 30 families; 0.159 vs. 0.017 murders), though the gap is smaller than for MP.

Economic, enforcement, and land-use characteristics differ correspondingly. GDP per capita is modestly higher in treated municipalities of both programs. Enforcement is far more intense in MP-treated municipalities (64 fines and R$ 65 million on average, against 9 fines and R$ 3.7 million in MP controls). MP-treated municipalities also have much larger cumulative protected area (5,600 ha vs. 1,071) and markedly lower population density (3.8 vs. 26.9 inhabitants/km$^2$), consistent with frontier locations; PMV-treated municipalities are denser than their controls. Registration is higher in treated municipalities of both programs (25.0% vs. 13.6% for MP; 18.1% vs. 8.0% for PMV).

Taken together, treatment—especially MP—is concentrated in systematically more conflict-prone municipalities with distinct enforcement and land-use profiles. This underscores the need to condition on pre-treatment trends rather than to rely on cross-sectional comparisons, as our SDID design does.

**Table A.1**
Summary statistics (MP, PMV, neither MP/PMV, and full sample).

| | Panel A | Panel B: MP | | Panel C: PMV | | Panel D: Neither MP nor PMV | |
|---|---|---|---|---|---|---|---|
| | Full | Control | Treated | Control | Treated | Control | Neither |
| Land Conflicts (Count) | 0.503 | 0.448 | 2.175 | 0.440 | 0.871 | 0.944 | 0.418 |
| | (1.599) | (1.503) | (2.960) | (1.564) | (1.741) | (1.857) | (1.530) |
| | [7584] | [7344] | [240] | [6468] | [1116] | [1224] | [6360] |
| Families in Conflict (Count) | 40.851 | 35.681 | 199.037 | 29.718 | 105.373 | 106.511 | 28.215 |
| | (174.675) | (165.215) | (321.673) | (132.777) | (316.806) | (307.939) | (130.985) |
| | [7584] | [7344] | [240] | [6468] | [1116] | [1224] | [6360] |
| Attempted Murders | 0.036 | 0.027 | 0.304 | 0.020 | 0.126 | 0.122 | 0.019 |
| | (0.385) | (0.330) | (1.137) | (0.295) | (0.703) | (0.680) | (0.294) |
| | [7584] | [7344] | [240] | [6468] | [1116] | [1224] | [6360] |
| Murders | 0.038 | 0.022 | 0.533 | 0.017 | 0.159 | 0.171 | 0.013 |
| | (0.301) | (0.195) | (1.206) | (0.166) | (0.663) | (0.667) | (0.137) |
| | [7584] | [7344] | [240] | [6468] | [1116] | [1224] | [6360] |
| Threats | 0.226 | 0.181 | 1.608 | 0.156 | 0.630 | 0.663 | 0.142 |
| | (1.303) | (1.182) | (3.002) | (1.180) | (1.812) | (1.900) | (1.135) |
| | [7584] | [7344] | [240] | [6468] | [1116] | [1224] | [6360] |
| GDP pc (R$2019) | 5776.905 | 5749.029 | 6628.129 | 5631.571 | 6620.125 | 6660.217 | 5607.043 |
| | (5,690.182) | (5,728.063) | (4,302.101) | (4,793.925) | (9,281.041) | (8,982.662) | (4,787.635) |
| | [6938] | [6718] | [220] | [5918] | [1020] | [1119] | [5819] |
| Fine Count | 11.124 | 9.390 | 64.192 | 9.481 | 20.647 | 25.090 | 8.436 |
| | (32.244) | (28.106) | (76.138) | (31.175) | (36.436) | (46.282) | (27.984) |
| | [7584] | [7344] | [240] | [6468] | [1116] | [1224] | [6360] |
| Total Fines (millions R$2019) | 5.680 | 3.741 | 65.020 | 3.078 | 20.758 | 20.771 | 2.775 |
| | (48.594) | (29.472) | (211.152) | (27.812) | (106.332) | (102.332) | (27.380) |
| | [7584] | [7344] | [240] | [6468] | [1116] | [1224] | [6360] |

(*continued on next page*)

**Table A.1** (*continued*)

| | Panel A | Panel B: MP | | Panel C: PMV | | Panel D: Neither MP nor PMV | |
|---|---|---|---|---|---|---|---|
| | Full | Control | Treated | Control | Treated | Control | Neither |
| Rainfall (mm) | 162.231 | 162.100 | 166.242 | 159.156 | 180.019 | 178.246 | 159.143 |
| | (46.037) | (46.382) | (33.644) | (46.709) | (37.269) | (38.209) | (46.774) |
| | [7572] | [7332] | [240] | [6456] | [1116] | [1224] | [6348] |
| Max Temp (K) | 316.239 | 316.228 | 316.561 | 316.307 | 315.841 | 315.835 | 316.316 |
| | (7.312) | (7.356) | (5.789) | (7.459) | (6.383) | (6.246) | (7.498) |
| | [7572] | [7332] | [240] | [6456] | [1116] | [1224] | [6348] |
| Cumulative protected area (ha) | 1214.385 | 1071.062 | 5600.077 | 989.351 | 2518.614 | 2598.003 | 948.104 |
| | (5,951.633) | (5,244.025) | (16,094.184) | (4,733.109) | (10,438.926) | (10,345.129) | (4,606.258) |
| | [7584] | [7344] | [240] | [6468] | [1116] | [1224] | [6360] |
| Pop. Density | 26.213 | 26.949 | 3.757 | 23.828 | 40.049 | 36.870 | 24.164 |
| | (133.986) | (136.098) | (3.715) | (95.284) | (263.159) | (251.450) | (96.055) |
| | [7569] | [7329] | [240] | [6456] | [1113] | [1221] | [6348] |
| CAR Area (ha) | 79,332.386 | 57,560.718 | 342,571.640 | 23,713.000 | 109,833.340 | 109,833.340 | 23,713.000 |
| | (212,555.020) | (137,959.550) | (538,251.899) | (84,604.910) | (251,877.972) | (251,877.972) | (84,604.910) |
| | [1008] | [931] | [77] | [357] | [651] | [651] | [357] |
| CAR Eligible Area (ha) | 432,094.560 | 360,998.900 | 1,291,705.700 | 209,737.230 | 554,032.440 | 554,032.440 | 209,737.230 |
| | (545,161.572) | (450,574.598) | (801,903.662) | (360,128.160) | (589,367.849) | (589,367.849) | (360,128.160) |
| | [1008] | [931] | [77] | [357] | [651] | [651] | [357] |
| CAR Property Size (ha) | 938.010 | 901.129 | 1383.929 | 854.066 | 984.044 | 984.044 | 854.066 |
| | (4,563.737) | (4,580.844) | (4,355.380) | (4,399.757) | (4,653.898) | (4,653.898) | (4,399.757) |
| | [1008] | [931] | [77] | [357] | [651] | [651] | [357] |
| CAR Registered (%) | 14.509 | 13.641 | 25.008 | 7.973 | 18.094 | 18.094 | 7.973 |
| | (21.073) | (20.062) | (28.939) | (12.039) | (23.921) | (23.921) | (12.039) |
| | [1008] | [931] | [77] | [357] | [651] | [651] | [357] |

*Notes:* Means with standard deviations in parentheses and observation counts in square brackets; the unit is the municipality–year, 2001–2012. CAR Area, Eligible Area, Property Size, and Registered % are available for the state of Pará only. MP = Priority Municipalities; PMV = Green Municipalities.

## Appendix B. CAR registration trends (2006–2012)

The CAR was piloted in Pará in 2006 and became a key component of Brazil's environmental enforcement architecture from 2008. Registration was initially voluntary but was increasingly tied to rural credit access, embargo avoidance, and participation in programs such as the PMV. Mato Grosso followed a distinct, more gradual trajectory, building on its earlier licensing-based registry.

Table B.2 summarizes CAR registration in both states from 2006 to 2012, reporting registered area (thousands of hectares) and the corresponding share of eligible area as reported in the source studies. The figures are digitized from prior studies (L'Roe et al., 2016; Jung et al., 2017; Alix-Garcia et al., 2018; Jung et al., 2022) and are consistent with official state reports.

**Table B.2**
Share of land registered in CAR by year in Pará and Mato Grosso.

| State | Year | CAR Registered Area (000 ha) | Share of Eligible Area (%) |
|---|---|---|---|
| Pará | 2006 | 3.34 | 0.01 |
| | 2007 | 35.15 | 0.03 |
| | 2008 | 614.92 | 0.73 |
| | 2009 | 5458.50 | 6.07 |
| | 2010 | 17,862.70 | 23.46 |
| | 2011 | 24,704.85 | 32.62 |
| | 2012 | 31,287.58 | 38.66 |
| Mato Grosso | 2006 | 17,366.08 | 21.75 |
| | 2007 | 20,416.45 | 25.26 |
| | 2008 | 24,718.77 | 30.81 |
| | 2009 | 27,393.62 | 34.30 |
| | 2010 | 32,824.24 | 41.15 |
| | 2011 | 37,108.42 | 46.74 |
| | 2012 | 39,802.14 | 50.24 |

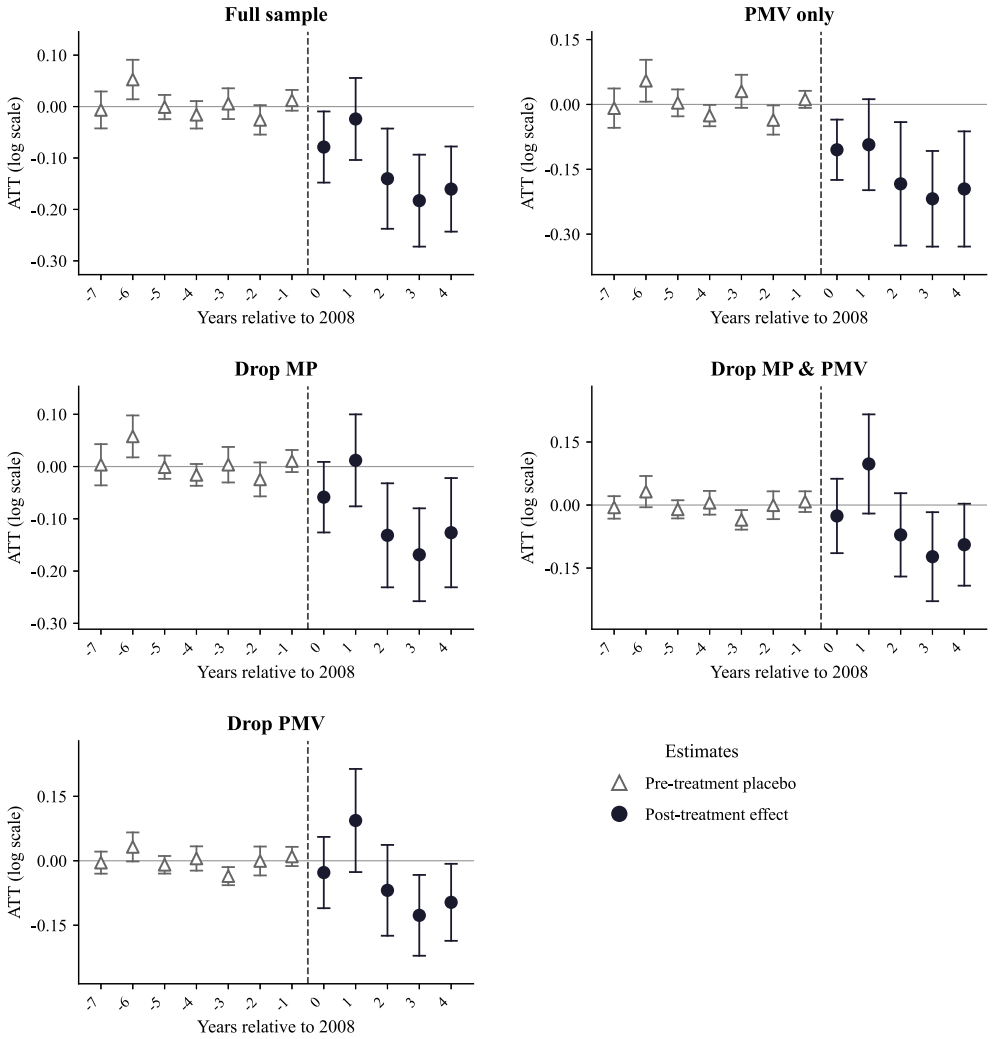
*Note:* Registered area in thousands of hectares; eligible area fixed at 62.2 million ha (Pará) and 73.7 million ha (Mato Grosso). Adoption was negligible before 2008 in Pará and 2009 in Mato Grosso, and was driven by policy incentives, credit conditioning, and enforcement. Data from (L'Roe et al., 2016; Jung et al., 2017; Alix-Garcia et al., 2018; Jung et al., 2022).

## Appendix C. Event study and SDID trend plots

This appendix presents two complementary sets of figures for Table 2: event-study plots (Section C.1), which show year-by-year treatment-effect estimates relative to 2008, and SDID-weighted outcome-trend plots (Section C.2), which show outcome levels for treated Pará municipalities alongside their synthetic control. The first assesses pre-treatment parallelism; the second assesses how well the synthetic control tracks Pará's pre-treatment levels.

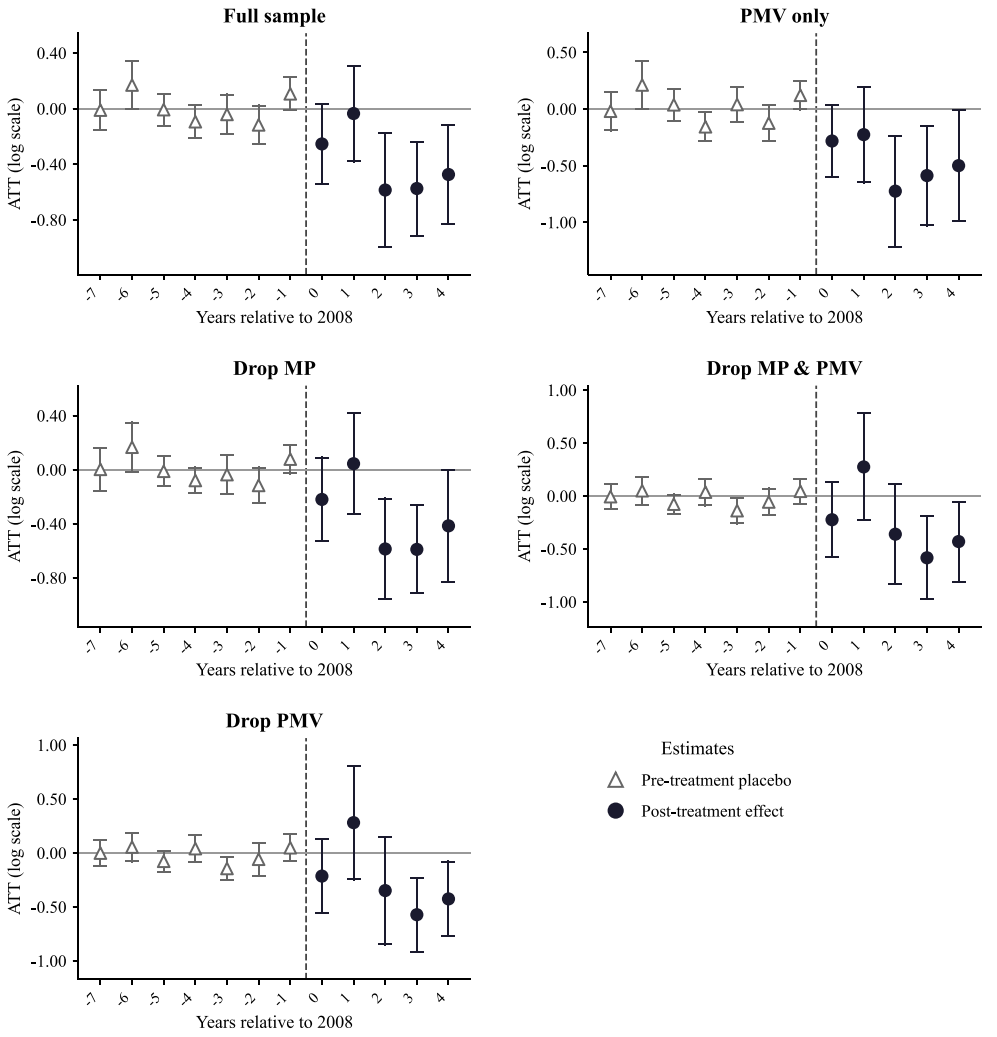
### *C.1. Event study plots*

Figs. C.1–C.5 report SDID event-study estimates for all five outcomes across five samples: full, PMV-only, Drop-MP, Drop-MP&PMV, and Drop-PMV. Each panel plots log-point ATTs by year relative to 2008; hollow triangles are pre-treatment placebo estimates, filled circles post-treatment estimates, and capped bars denote 95% bootstrap confidence intervals (1,000 replications). Pre-treatment estimates are flat and indistinguishable from zero across all outcomes and subsamples, supporting the parallel-trends assumption.



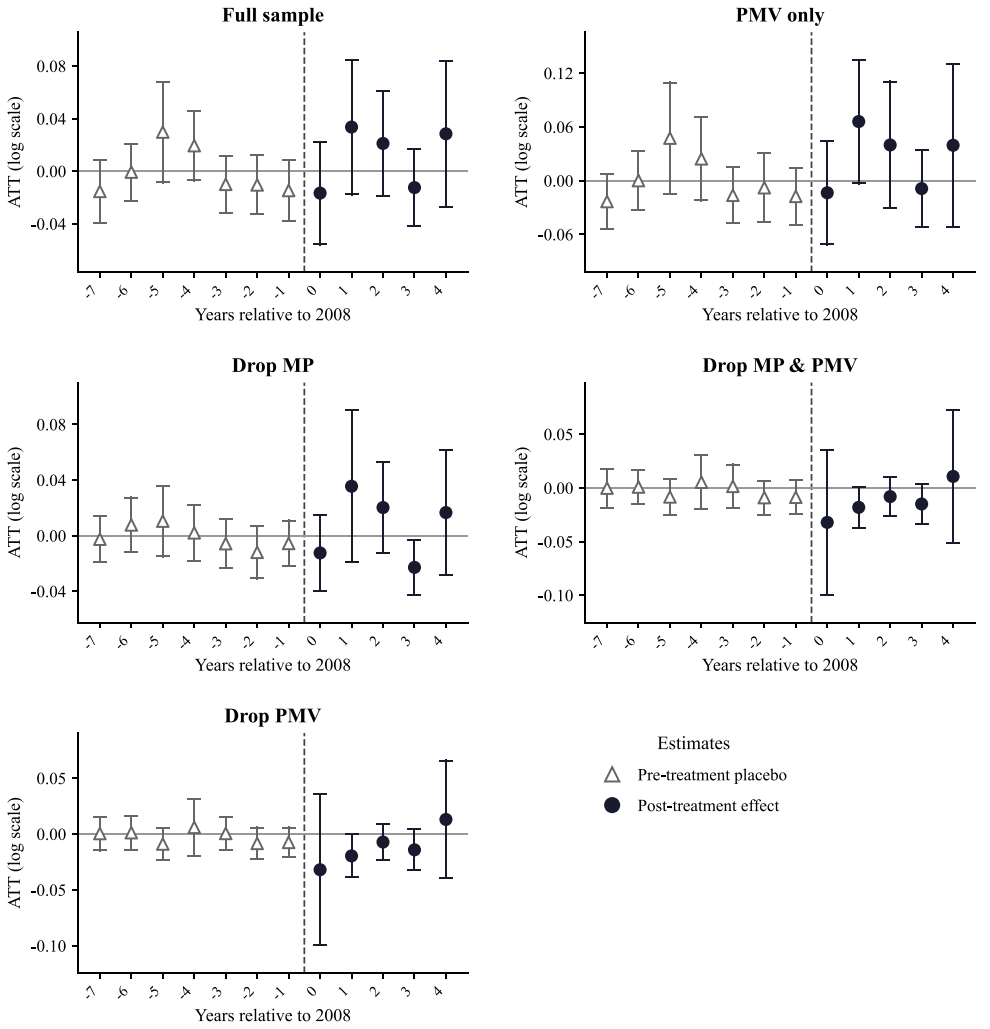

Notes: SDID event-study ATTs, conflict events. 95% bootstrap CI (1000 reps). Triangles = pre-treatment placebo. Circles = post-treatment effect. Dashed = 2007.5.

**Fig. C.1.** Event Study: Total land conflict Events. Each panel shows SDID ATT estimates for one subsample. Hollow triangles = pre-treatment placebo (years −7 to −1); filled circles = post-treatment effect (years 0 to +4); Capped bars = 95% bootstrap CI (1,000 replications); Dashed line = 2007.5.



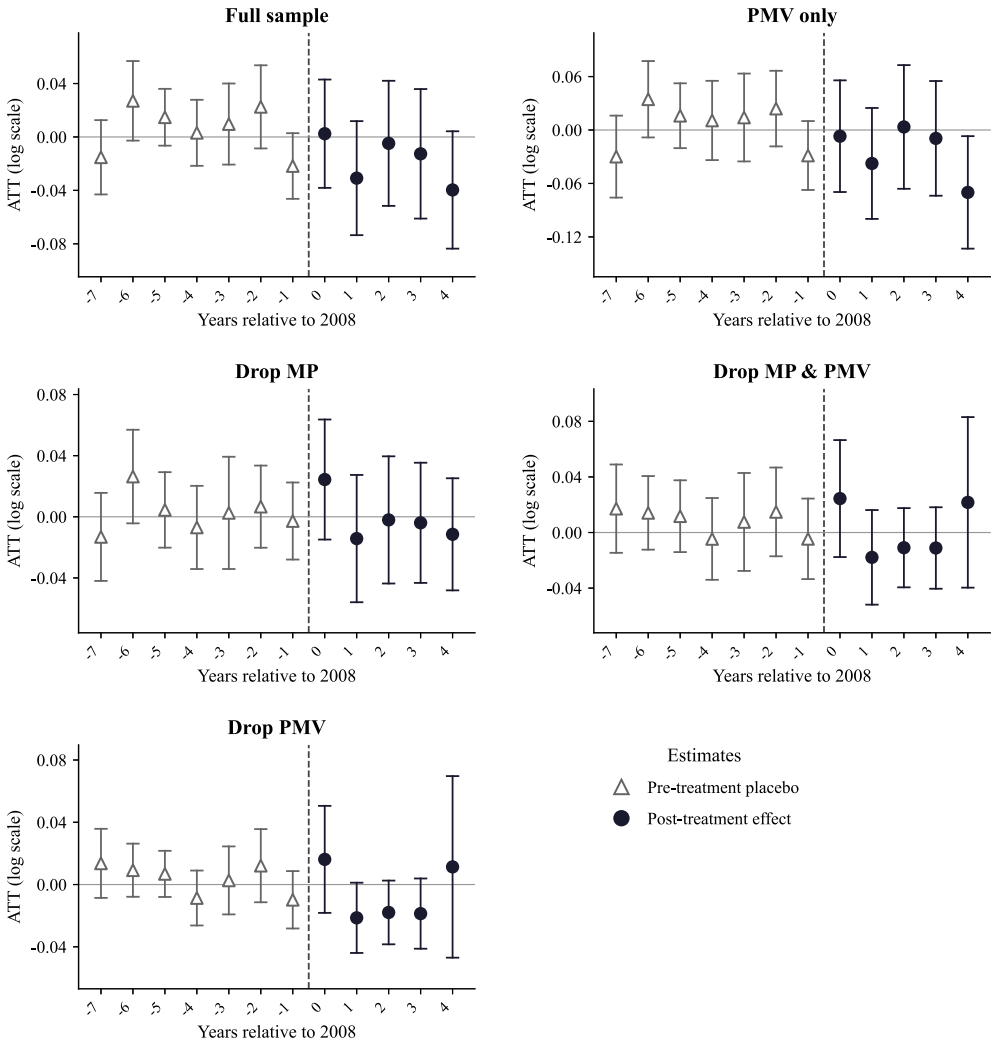

Notes: SDID event-study ATTs, families affected. 95% bootstrap CI (1000 reps). Triangles = pre-treatment placebo. Circles = post-treatment effect. Dashed = 2007.5.

**Fig. C.2.** Event Study: Families affected by land conflicts. Panels, markers, and intervals as in Fig. C.1.



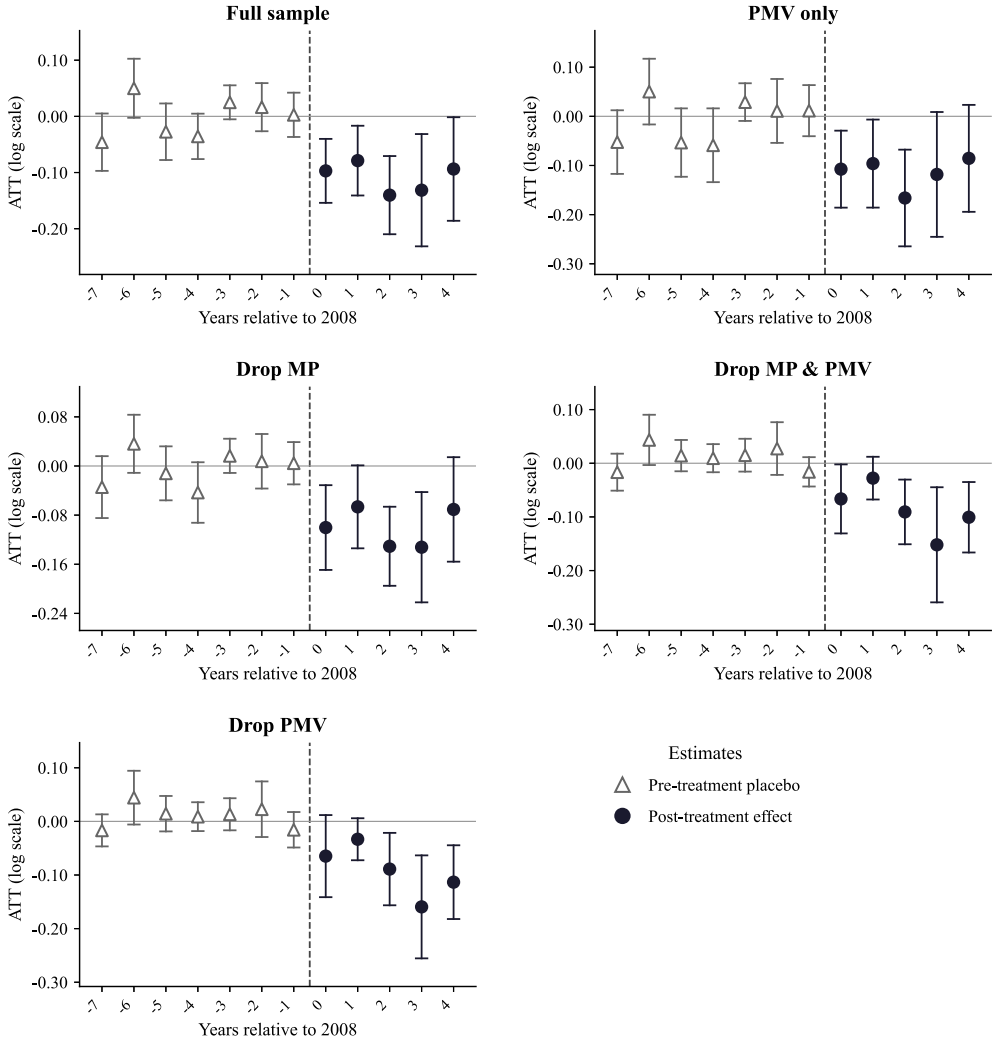

Notes: SDID event-study ATTs, attempted murders. 95% bootstrap CI (1000 reps). Triangles = pre-treatment placebo. Circles = post-treatment effect. Dashed = 2007.5.

**Fig. C.3.** Event Study: Attempted Murders. Panels, markers, and intervals as in Fig. C.1.




Notes: SDID event-study ATTs, murders. 95% bootstrap CI (1000 reps). Triangles = pre-treatment placebo. Circles = post-treatment effect. Dashed = 2007.5.

**Fig. C.4.** Event Study: Murders. Panels, markers, and intervals as in Fig. C.1.



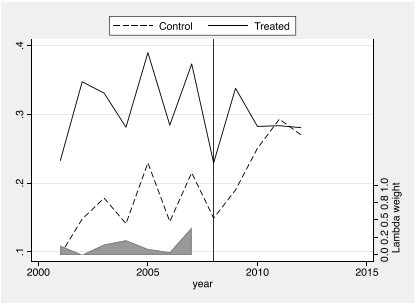

Notes: SDID event-study ATTs, death threats. 95% bootstrap CI (1000 reps). Triangles = pre-treatment placebo. Circles = post-treatment effect. Dashed = 2007.5.

**Fig. C.5.** Event Study: Death Threats. Panels, markers, and intervals as in Fig. C.1.

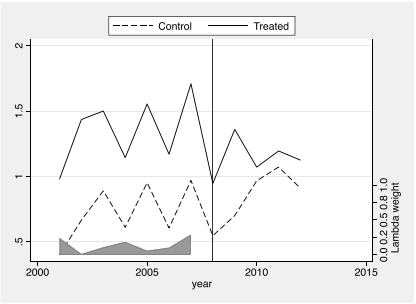
### *C.2. SDID outcome trend plots*

Fig. C.6 displays SDID-weighted outcome paths for all five outcomes (full sample). For each outcome the solid line is the average path for treated Pará municipalities and the dashed line is the SDID-weighted synthetic control; lambda weights on the right axis, concentrated in the years just before 2008, show the time weights used in matching. Following Arkhangelsky et al. (2021) (whose Fig. 1 presents an analogous plot for California cigarette sales), these show outcome *levels* rather than year-by-year effects, allowing visual assessment of pre-treatment fit and post-treatment divergence.

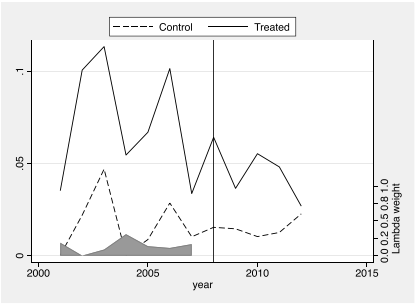
Pre-treatment alignment is close for all five outcomes: treated and synthetic paths track together from 2001 to 2007 with no systematic divergence, confirming that the synthetic control matches Pará's pre-treatment trajectory. For conflict events, families affected, murders, and death threats, the treated municipalities lie at or above their synthetic control throughout; after 2008 the gap narrows relative to the pre-treatment period, consistent with the negative ATTs. Because the ATT is a difference-in-differences, a negative effect implies this post-2008 narrowing of the gap—not the treated line crossing below the synthetic control. The narrowing is most pronounced for families affected, consistent with the largest ATT (−33.4%). For attempted violence the two paths remain close with no systematic narrowing, consistent with its small, imprecise ATT.



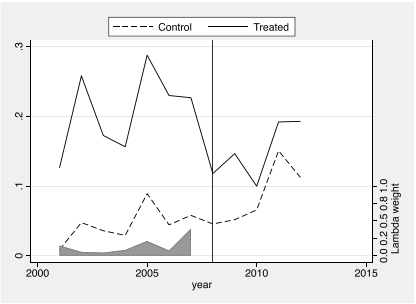

(a) Land conflict events



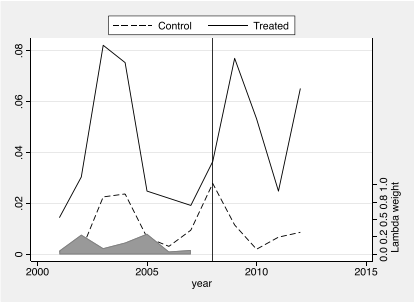

(b) Families affected




(c) Murders




(d) Death threats




(e) Attempted violence

**Fig. C.6.** SDID-weighted outcome trend plots (full sample). Solid line = treated Pará municipalities; Dashed line = SDID-weighted synthetic control; vertical Dashed line = 2008. Lambda weights on the right axis are concentrated just before 2008. All outcomes are $\log(1 + y)$. Close pre-treatment alignment confirms the synthetic match; the post-2008 narrowing of the treated–synthetic gap corroborates the ATTs in Table 2. Generated with the `sdid` command (Clarke et al., 2024) using default (non-projected) covariates for visualization; main ATTs use the projected specification.

## Appendix D. Robustness to lagged covariates

To address simultaneity between treatment and time-varying covariates, we re-estimate the main specification with one-year lagged covariates. The main specification uses contemporaneous controls to preserve the full 2001–2012 panel; the lagged specification drops 2001 and ensures that treatment cannot mechanically affect covariates. Both include the same five controls. Table D.3 reports the two side by side.

Both specifications are qualitatively and quantitatively consistent: the ATTs remain negative and significant for conflict events, families affected, murders, and threats. The lagged estimates are slightly larger, especially for murders and threats, indicating that contemporaneous controls do not bias the main results upward and that the main specification is, if anything, conservative.

**Table D.3**
Impact of CAR implementation on Land Conflict: Baseline vs. Lagged Covariates.

| | (1) Conflict | (2) Families | (3) Attempts | (4) Murders | (5) Threats |
|---|---|---|---|---|---|
| Panel A: Baseline model (2001–2012) | | | | | |
| ATT Estimate | −0.122*** | −0.407*** | 0.010* | −0.019*** | −0.106*** |
| Standard Error | (0.030) | (0.113) | (0.006) | (0.006) | (0.021) |
| Pre-Treatment Mean (log) | 0.172 | 0.745 | 0.009 | 0.008 | 0.039 |
| Pre-Treatment Mean (level) | 0.331 | 28.561 | 0.016 | 0.012 | 0.118 |
| Relative Effect (%) | −11.5 | −33.4 | +1.0 | −1.9 | −10.0 |
| Observations | 7584 | 7584 | 7584 | 7584 | 7584 |
| Municipalities | 632 | 632 | 632 | 632 | 632 |
| Panel B: Lagged covariates (excluding 2001) | | | | | |
| ATT Estimate | −0.122*** | −0.415*** | 0.008 | −0.023*** | −0.115*** |
| Standard Error | (0.030) | (0.121) | (0.006) | (0.006) | (0.020) |
| Pre-Treatment Mean (log) | 0.186 | 0.809 | 0.009 | 0.009 | 0.040 |
| Pre-Treatment Mean (level) | 0.362 | 31.431 | 0.017 | 0.014 | 0.115 |
| Relative Effect (%) | −11.5 | −34.0 | +0.8 | −2.3 | −10.9 |
| Observations | 6952 | 6952 | 6952 | 6952 | 6952 |
| Municipalities | 632 | 632 | 632 | 632 | 632 |

*Notes:* SDID estimates with log-transformed outcomes. Panel A uses the full sample (treated = Pará; controls = Legal Amazon excluding Mato Grosso); Panel B uses the same sample excluding 2001, with one-year lagged covariates. Relative effects are $(e^{\text{ATT}} - 1) \times 100$. All models include municipality and year fixed effects and control for GDP per capita, precipitation, temperature, cumulative protected area, and population density. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

## Appendix E. Alternative outcome specifications

Our main results use SDID with log-transformed outcomes. To show they are not an artifact of how we handle the many zeros, this appendix reports specifications that avoid the log transformation: raw counts under linear and Poisson QMLE estimators; the inverse hyperbolic sine (IHS) transformation; and linear probability models for the binary extensive margin. All use the Wooldridge (2021, 2023) `jwdid` staggered DiD estimator (`hettype(twfe)`) on the same balanced panel of 632 municipalities, with ATTs recovered in natural units via `estat simple`. The count specification estimates

$$\mathbb{E}[Y_{it} \mid X_{it-1},\, D_{it}] \;=\; \exp\Big(\alpha_i + \lambda_t + \beta\, D_{it} + \gamma\, X_{it-1}\Big), \tag{E.1}$$

where $Y_{it}$ is the raw count, $D_{it}$ the CAR indicator, $X_{it-1}$ one-year lagged covariates, and $\alpha_i$, $\lambda_t$ are municipality and year fixed effects, with standard errors clustered by municipality. Because this specification uses neither the SDID unit nor time weights, the comparison with the main estimates is an independent cross-check.

### *E.1. Raw count outcomes: linear and poisson estimates*

Table E.4 reports estimates on raw counts. Columns (1)–(2) impose no controls (linear and Poisson), allowing a direct functional-form comparison; columns (3)–(4) add the full control set in unlogged and logged form; columns (5)–(8) add each control individually to the Poisson specification. Three patterns emerge. The linear and Poisson no-covariate estimates are numerically close for every outcome. Signs are uniformly negative except for attempted violence, which is small, positive, and never significant—as in the main results. Families affected and threats are the most robust: families is significant at 5% in six of eight columns, and threats are significant at 1% in three columns and 5% in two more. Poisson QMLE faces quasi-complete separation for the sparsest outcomes when a single continuous covariate is the only predictor (columns 5 and 8); the linear estimates in column (1), unaffected by sparsity, are our primary count-outcome reference.

**Table E.4**
CAR implementation and Land Conflict: Raw Count Outcomes.

| | *No covariates* | | *Full covariate set* | | *Poisson: sequential covariate addition* | | | |
|---|---|---|---|---|---|---|---|---|
| | (1) Linear | (2) Poisson | (3) Linear unlogged | (4) Linear logged | (5) + GDP per capita | (6) + Pop. density | (7) + Precip. & temp. | (8) + Protected area |
| **Panel A: Conflict events** Pre-treatment mean = 0.331 | | | | | | | | |
| ATT | −0.233 | −0.029 | −0.320** | −0.269* | −0.413* | −0.533*** | −0.300 | −1.025 |
| SE | (0.159) | (0.150) | (0.158) | (0.158) | (0.223) | (0.201) | (0.186) | (0.677) |
| *N* | 7584 | 7584 | 7584 | 7584 | 7584 | 7584 | 7584 | 7584 |
| Mun. | 632 | 632 | 632 | 632 | 632 | 632 | 632 | 632 |
| **Panel B: Families affected** Pre-treatment mean = 28.561 | | | | | | | | |
| ATT | −30.983** | −28.310** | −31.736* | −43.156** | −77.582 | −40.498* | −17.478 | −37.459* |
| SE | (13.645) | (11.957) | (18.307) | (17.324) | (138.538) | (22.252) | (18.164) | (22.585) |
| *N* | 7584 | 7584 | 7584 | 7584 | 7584 | 7584 | 7584 | 7584 |
| Mun. | 632 | 632 | 632 | 632 | 632 | 632 | 632 | 632 |
| **Panel C: Attempted violence** Pre-treatment mean = 0.016 | | | | | | | | |
| ATT | 0.038 | 0.044 | 0.036 | 0.053 | −3.182 | 0.005 | 0.015 | —[a] |
| SE | (0.030) | (0.029) | (0.053) | (0.049) | (20.407) | (0.046) | (0.048) | |
| *N* | 7584 | 7584 | 7584 | 7584 | 7584 | 7584 | 7584 | — |
| Mun. | 632 | 632 | 632 | 632 | 632 | 632 | 632 | — |
| **Panel D: Murders** Pre-treatment mean = 0.012 | | | | | | | | |
| ATT | −0.069** | −0.059** | −0.083** | −0.067 | −72.048 | −0.145* | −0.128 | —[a] |
| SE | (0.030) | (0.030) | (0.041) | (0.041) | (335.140) | (0.078) | (0.087) | |
| *N* | 7584 | 7584 | 7584 | 7584 | 7584 | 7584 | 7584 | — |
| Mun. | 632 | 632 | 632 | 632 | 632 | 632 | 632 | — |
| **Panel E: Threats** Pre-treatment mean = 0.118 | | | | | | | | |
| ATT | −0.306*** | −0.230*** | −0.218 | −0.142 | −0.673 | −0.673** | −0.550** | —[a] |
| SE | (0.089) | (0.083) | (0.148) | (0.142) | (0.450) | (0.275) | (0.249) | |
| *N* | 7584 | 7584 | 7584 | 7584 | 7584 | 7584 | 7584 | — |
| Mun. | 632 | 632 | 632 | 632 | 632 | 632 | 632 | — |
| *Specification* | | | | | | | | |
| Estimator | OLS | PQMLE | OLS | OLS | PQMLE | PQMLE | PQMLE | PQMLE |
| Municipality FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Year FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| State FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| GDP per capita | × | × | ✓ | ✓ | ✓ | × | × | × |
| Population density | × | × | ✓ | ✓ | × | ✓ | × | × |
| Precip. & temp. | × | × | ✓ | ✓ | × | × | ✓ | × |
| Protected area | × | × | ✓ | ✓ | × | × | × | ✓ |
| Covariate form | — | — | Unlogged | Logged | Unlogged | Unlogged | Unlogged | Unlogged |

*Notes:* Dependent variables are raw counts (no transformation). All columns use the Wooldridge (2023) `jwdid` estimator (`hettype(twfe)`); ATTs are marginal effects in natural units via `estat simple`. Columns (1)–(2) use no controls; (3)–(4) add the full set unlogged/logged; (5)–(8) add each control individually to the Poisson specification. Pre-treatment means are for control-state municipalities, 2001–2007. Standard errors clustered by municipality.

[a] The Poisson log-likelihood did not converge for attempted violence, murders, and threats in column (8): with pre-treatment means below 0.02 and cumulative protected area as the sole covariate, zero-inflation causes quasi-complete separation. The linear estimates in column (1) are consistent regardless of sparsity. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

### *E.2. IHS transformation and extensive margin*

Table E.5 reports two further specifications. Panel A uses the IHS transformation, $\text{IHS}(y) = \ln(y + \sqrt{y^2 + 1})$, defined at zero and approximating $\ln(y)$ for large values (Bellemare and Wichman, 2020); coefficients are approximate log-point changes. Panel B uses a linear probability model on an indicator for any event (extensive margin), with ATTs in probability units. The IHS estimates replicate the count pattern: without controls, conflict events, families affected, murders, and threats imply reductions of roughly 13%, 37%, 4%, and 13%, with conflict events and threats remaining significant under logged covariates. The extensive-margin estimates show that the effect operates substantially through incidence: the probability of any conflict event falls by 7.6 percentage points and of any threat by 7.4 points; threats remain significant at 1% in both specifications, the most robust outcome across this appendix. Together the two panels indicate that CAR reduced both the incidence and the severity of conflict.

**Table E.5**
CAR implementation and land conflict: IHS and extensive margin estimates.

| | Panel A: IHS outcomes | | Panel B: Extensive margin (LPM) | |
|---|---|---|---|---|
| | (1) No covs. | (2) Logged covs. | (3) No covs. | (4) Logged covs. |
| *Conflict events* IHS pre-mean = 0.325; binary pre-mean = 0.182 | | | | |
| ATT | −0.137*** | −0.123** | −0.076*** | −0.070* |
| SE | (0.038) | (0.051) | (0.024) | (0.036) |
| *N* | 7584 | 7584 | 7584 | 7584 |
| Mun. | 632 | 632 | 632 | 632 |
| *Families affected* IHS pre-mean = 4.049; binary pre-mean = 0.174 | | | | |
| ATT | −0.369*** | −0.218 | −0.072*** | −0.050 |
| SE | (0.127) | (0.189) | (0.024) | (0.035) |
| *N* | 7584 | 7584 | 7584 | 7584 |
| Mun. | 632 | 632 | 632 | 632 |
| *Attempted violence* IHS pre-mean = 0.016; binary pre-mean = 0.009 | | | | |
| ATT | 0.013 | 0.019 | 0.002 | 0.009 |
| SE | (0.014) | (0.024) | (0.010) | (0.017) |
| *N* | 7584 | 7584 | 7584 | 7584 |
| Mun. | 632 | 632 | 632 | 632 |
| *Murders* IHS pre-mean = 0.012; binary pre-mean = 0.009 | | | | |
| ATT | −0.040** | −0.042 | −0.028** | −0.031 |
| SE | (0.017) | (0.030) | (0.013) | (0.027) |
| *N* | 7584 | 7584 | 7584 | 7584 |
| Mun. | 632 | 632 | 632 | 632 |
| *Threats* IHS pre-mean = 0.118; binary pre-mean = 0.034 | | | | |
| ATT | −0.131*** | −0.091* | −0.074*** | −0.075*** |
| SE | (0.033) | (0.048) | (0.018) | (0.028) |
| *N* | 7584 | 7584 | 7584 | 7584 |
| Mun. | 632 | 632 | 632 | 632 |
| *Specification* | | | | |
| Estimator | OLS | OLS | OLS | OLS |
| Municipality FE | ✓ | ✓ | ✓ | ✓ |
| Year FE | ✓ | ✓ | ✓ | ✓ |
| State FE | ✓ | ✓ | ✓ | ✓ |
| Logged covariates | × | ✓ | × | ✓ |

*Notes:* All specifications use the Wooldridge (2023) `jwdid` estimator (`hettype(twfe)`), ATTs via `estat simple`. Panel A uses the IHS transformation $\text{IHS}(y) = \ln(y + \sqrt{y^2 + 1})$; coefficients are approximate log-point changes. Panel B uses an indicator for any event; ATTs are probability-unit changes (e.g., −0.076 implies a 7.6 percentage-point reduction in the probability of any conflict event). Pre-treatment means are for control-state municipalities, 2001–2007. Logged covariates: GDP per capita, precipitation, maximum temperature, cumulative protected area, and population density. Standard errors are clustered by municipality. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

### *E.3. SDID with count and logged outcomes*

Table E.6 reports SDID estimates on raw-count outcomes (Panel A) and on the log-transformed outcomes of the main results (Panel B), using the same lagged logged covariate set and estimator, so differences reflect the outcome transformation only. In Panel A the ATT on conflict events is −0.286 incidents per municipality-year (pre-treatment mean 0.331), consistent with the linear estimate in Table E.4; families affected fall by about 33 per municipality-year against a mean of 28.6. The pattern of signs is identical across the SDID level estimator and every Wooldridge-based specification.

The Panel A percentage effects should be read with caution: because pre-treatment means for attempted violence and murders are near zero, level-metric percentages amplify mechanically (e.g., −312% for murders). These are a mechanical artefacts and are *not* reported or interpreted as headline effects anywhere in the paper; for these outcomes we rely on the absolute ATTs and on the extensive-margin estimates in Tables 2 and E.5. The two panels' percentage metrics also differ by construction—Panel A uses $\widehat{\text{ATT}}/\bar{y}_{\text{pre}}$ while Panel B uses $(e^{\widehat{\text{ATT}}} - 1)$—and the SDID re-estimates its weights separately for level and logged outcomes, so the panels are complementary rather than a transformation of one another: Panel A gives interpretable absolute magnitudes and Panel B the semi-elasticities that are standard in the literature.

**Table E.6**
SDID Estimates: Count and Logged Outcomes.

| | (1) Conflict | (2) Families | (3) Attempts | (4) Murders | (5) Threats |
|---|---|---|---|---|---|
| Panel A: Count dependent variable | | | | | |
| ATT Estimate | −0.286*** | −32.812*** | 0.029* | −0.038*** | −0.378*** |
| Standard Error | (0.102) | (8.230) | (0.016) | (0.011) | (0.069) |
| Pre-Treatment Mean | 0.331 | 28.561 | 0.016 | 0.012 | 0.118 |
| Relative Effect (%) | −86.3 | −114.9 | +182.3 | −312.2 | −321.1 |
| Panel B: Log dependent variable (main specification) | | | | | |
| ATT Estimate | −0.122*** | −0.407*** | 0.010* | −0.019*** | −0.106*** |
| Standard Error | (0.030) | (0.113) | (0.006) | (0.006) | (0.021) |
| Pre-Treatment Mean | 0.172 | 0.745 | 0.009 | 0.008 | 0.039 |
| Relative Effect (%) | −11.5 | −33.4 | +1.0 | −1.9 | −10.0 |
| Observations | 7584 | 7584 | 7584 | 7584 | 7584 |
| Municipalities | 632 | 632 | 632 | 632 | 632 |

*Notes:* SDID estimates (Arkhangelsky et al., 2021); treatment equals one from 2008 onward in Pará. Both panels use the same logged covariates (GDP per capita, precipitation, maximum temperature, cumulative protected area, population density) via the projected method (Clarke et al., 2024). Panel A uses raw counts (ATTs are absolute changes; relative effects $100 \times \widehat{\text{ATT}}/\bar{y}_{\text{pre}}$); Panel B uses logs (relative effects $(e^{\widehat{\text{ATT}}} - 1) \times 100$). Pre-treatment means are for control-state municipalities, 2001–2007 (logs in Panel B). Panel A percentages are unreliable for near-zero-mean outcomes. Standard errors are from placebo variance, 100 replications, `seed(2025)`. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

### *E.4. Extensive margin across subsamples*

Table E.5 reports the extensive margin for the full sample only. Because Table 2 reports the intensive-margin SDID estimates across five subsamples, we re-estimate the extensive-margin LPM on the same five subsamples so that the incidence margin can be read against the intensive margin sample-by-sample (Table E.7). Each cell is a separate Wooldridge (2023) `jwdid` LPM on $\mathbf{1}\{y > 0\}$ with the identical specification and subsample definitions used elsewhere; the full-sample column reproduces Table E.5, Panel B. The pattern tracks the intensive margin: reductions in the probability of any conflict event or affected family are strongest in the full, Drop-MP, and PMV-only samples and attenuate once both programs are removed, whereas the probability of any death threat falls at the 1% level in every subsample—the outcome most robust in the main table.

**Table E.7**
Extensive margin (LPM) across Subsamples.

| | Full Sample | Drop-MP | Drop-MP&PMV | Drop-PMV | PMV only |
|---|---|---|---|---|---|
| Panel A: No covariates | | | | | |
| *Conflict events* | | | | | |
| ATT | −0.076*** | −0.066*** | −0.034 | −0.036 | −0.097*** |
| SE | (0.024) | (0.025) | (0.036) | (0.036) | (0.029) |
| *Families affected* | | | | | |
| ATT | −0.072*** | −0.063** | −0.035 | −0.037 | −0.092*** |
| SE | (0.024) | (0.025) | (0.037) | (0.037) | (0.029) |
| *Attempted violence* | | | | | |
| ATT | −0.002 | −0.000 | −0.010 | −0.010 | −0.009 |
| SE | (0.010) | (0.009) | (0.007) | (0.007) | (0.015) |
| *Murders* | | | | | |
| ATT | −0.028** | −0.005 | −0.014 | −0.016 | −0.035* |
| SE | (0.013) | (0.011) | (0.012) | (0.012) | (0.020) |
| *Threats* | | | | | |
| ATT | −0.074*** | −0.071*** | −0.083*** | −0.084*** | −0.069*** |
| SE | (0.018) | (0.017) | (0.023) | (0.023) | (0.023) |

(*continued on next page*)

**Table E.7** (*continued*)

| | Full Sample | Drop-MP | Drop-MP&PMV | Drop-PMV | PMV only |
|---|---|---|---|---|---|
| Panel B: Logged covariates | | | | | |
| *Conflict events* | | | | | |
| ATT | −0.070* | −0.061* | −0.083 | −0.082 | −0.063 |
| SE | (0.036) | (0.037) | (0.052) | (0.052) | (0.040) |
| *Families affected* | | | | | |
| ATT | −0.050 | −0.041 | −0.093* | −0.091* | −0.034 |
| SE | (0.035) | (0.036) | (0.054) | (0.053) | (0.038) |
| *Attempted violence* | | | | | |
| ATT | −0.009 | −0.021 | −0.017 | −0.018 | −0.018 |
| SE | (0.017) | (0.015) | (0.019) | (0.019) | (0.024) |
| *Murders* | | | | | |
| ATT | −0.031 | −0.001 | −0.010 | −0.007 | −0.045 |
| SE | (0.027) | (0.025) | (0.018) | (0.019) | (0.036) |
| *Threats* | | | | | |
| ATT | −0.075*** | −0.070** | −0.099*** | −0.098*** | −0.070** |
| SE | (0.028) | (0.028) | (0.033) | (0.033) | (0.035) |
| Observations | 7584 | 7344 | 6360 | 6468 | 6972 |
| Municipalities | 632 | 612 | 530 | 539 | 581 |

*Notes:* Each cell reports the ATT from a separate Wooldridge (2023) `jwdid` LPM (`hettype(twfe)`), ATT via `estat simple`, on a binary indicator for any event (probability units; −0.076 implies a 7.6 percentage-point reduction). All models include municipality, year, and state fixed effects; Panel B adds the logged covariate set. Standard errors clustered by municipality. Subsamples are defined as in Table 2. The full-sample column reproduces Table E.5, Panel B. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

## Appendix F. Main results using projected and default options

Our main results use the projected covariate specification, which removes covariate influence from the outcome before constructing synthetic-control weights so that treatment effects are orthogonal to the included covariates (Arkhangelsky et al., 2021; Kranz, 2022; Clarke et al., 2024). Table F.8 re-estimates the models under the default (unprojected) specification, which lets covariates influence weight construction. The two are nearly identical: conflict events and families affected are −11.5% and −33.4% under projection and −11.5% and −33.6% under the default, with the smaller outcomes shifting negligibly (e.g., +1.0% to +0.8% for attempts; −1.9% to −1.8% for murders). This stability confirms the results are not artefacts of covariate handling.

**Table F.8**
Impact of CAR implementation on land conflict (2001–2012).

| | (1) Conflict | (2) Families | (3) Attempts | (4) Murders | (5) Threats |
|---|---|---|---|---|---|
| Panel A: Projected covariates | | | | | |
| ATT Estimate | −0.122*** | −0.407*** | 0.010* | −0.019*** | −0.106*** |
| Standard Error | (0.030) | (0.113) | (0.006) | (0.006) | (0.021) |
| Pre-Treatment Mean (log) | 0.172 | 0.745 | 0.009 | 0.008 | 0.039 |
| Pre-Treatment Mean (level) | 0.331 | 28.561 | 0.016 | 0.012 | 0.118 |
| Relative Effect (%) | −11.5 | −33.4 | +1.0 | −1.9 | −10.0 |
| Panel B: Default (unprojected) covariates | | | | | |
| ATT Estimate | −0.123*** | −0.409*** | 0.008 | −0.018*** | −0.104*** |
| Standard Error | (0.030) | (0.112) | (0.006) | (0.006) | (0.021) |
| Pre-Treatment Mean (log) | 0.172 | 0.745 | 0.009 | 0.008 | 0.039 |
| Pre-Treatment Mean (level) | 0.331 | 28.561 | 0.016 | 0.012 | 0.118 |
| Relative Effect (%) | −11.5 | −33.6 | +0.8 | −1.8 | −9.9 |
| Observations | 7584 | 7584 | 7584 | 7584 | 7584 |
| Municipalities | 632 | 632 | 632 | 632 | 632 |

*Notes:* SDID estimates with log-transformed outcomes; panels differ only in whether covariates are projected out before weight construction (Arkhangelsky et al., 2021; Kranz, 2022; Clarke et al., 2024). Relative effects are $(e^{\mathrm{ATT}} - 1) \times 100$. All models include municipality and year fixed effects and control for GDP per capita, precipitation, temperature, cumulative protected area, and population density. Pre-treatment (level) means are exponentiated control-group means before 2008. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

## Appendix G. Border-based regression discontinuity and DID at the pará state boundary

**Design.** We implement a geographic RD using signed distance from each municipality's centroid to the Pará boundary as the running variable (positive inside Pará). This compares municipalities just inside and just outside the border, plausibly exposed to the same regional shocks but differing in CAR exposure after 2008. We estimate local-linear models with a triangular kernel and an MSE-optimal bandwidth (`mserd`), controlling for log GDP per capita, precipitation, temperature, cumulative protected area, and population density, for the pooled period (2001–2012) and the pre- and post-2008 subperiods. Fig. G.7 shows the spatial distribution of municipalities.

**Manipulation test.** The McCrary (2008) density test shows a small, insignificant discontinuity at the boundary (conventional $p = 0.95$; robust $p = 0.60$), and binomial tests across bandwidths show no imbalance (Fig. G.8). We caution, however, that a density test is designed for settings where units sort across a cutoff on a manipulable running variable; here the running variable is fixed geography and municipalities cannot relocate, so the test speaks to compositional balance rather than behavioural sorting and is of limited relevance. We place primary weight on the border-band SDID of Section G.1, which uses municipalities directly across the border as SDID-weighted controls.

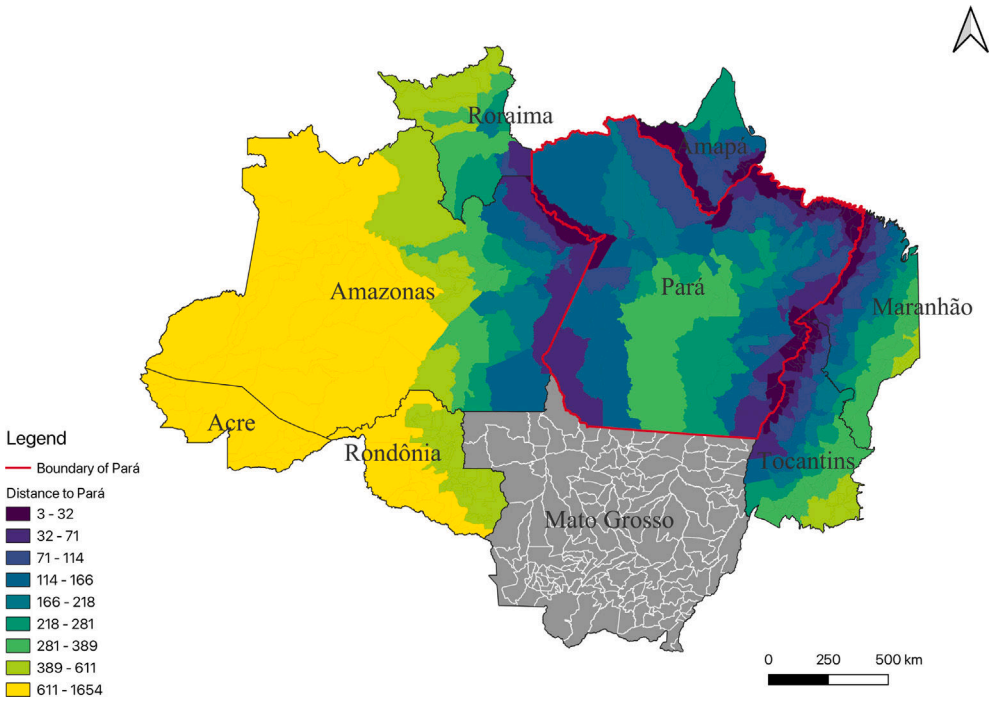



**Fig. G.7.** Distance from the Pará boundary (km), for municipality centroids across the Brazilian Amazon.

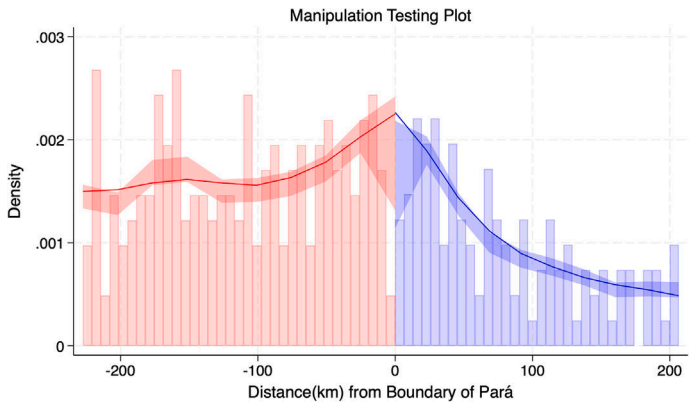



**Fig. G.8.** McCrary (2008) density test for the running variable.

**Table G.9**
Border RD estimates at the Pará boundary: pooled, pre-2008, and post-2008.

| | (1) Conflict | (2) Families | (3) Attempts | (4) Murders | (5) Threats |
|---|---|---|---|---|---|
| Pooled sample (2001–2012) | | | | | |
| Bias-corrected estimate | −0.0648 | −0.873** | −0.0444*** | 0.0108 | 0.0312 |
| | (0.107) | (0.440) | (0.0150) | (0.0131) | (0.0413) |
| Robust estimate | −0.0648 | −0.873 | −0.0444 | 0.0108 | 0.0312 |
| | (0.230) | (0.934) | (0.0308) | (0.0230) | (0.0569) |
| Control Mean | 0.503 | 40.851 | 0.036 | 0.038 | 0.226 |
| Optimal bandwidth | 36.66 | 35.52 | 77.81 | 66.27 | 56.22 |
| Observations | 7584 | 7584 | 7584 | 7584 | 7584 |
| Before 2008 (pre-implementation) | | | | | |
| Bias-corrected estimate | −0.0723 | −1.130** | −0.0369** | −0.000896 | 0.0598 |
| | (0.122) | (0.558) | (0.0152) | (0.0161) | (0.0536) |
| Robust estimate | −0.0723 | −1.130 | −0.0369 | −0.000896 | 0.0598 |
| | (0.269) | (1.195) | (0.0308) | (0.0307) | (0.0847) |
| Control Mean | 0.503 | 40.851 | 0.036 | 0.038 | 0.226 |
| Optimal bandwidth | 37.18 | 37.49 | 89.30 | 62.10 | 66.12 |
| Observations | 4424 | 4424 | 4424 | 4424 | 4424 |
| After 2008 (post-implementation) | | | | | |
| Bias-corrected estimate | −0.129 | −0.945** | −0.0603*** | 0.0153* | −0.0348 |
| | (0.136) | (0.463) | (0.0217) | (0.00839) | (0.0350) |
| Robust estimate | −0.129 | −0.945 | −0.0603* | 0.0153 | −0.0348 |
| | (0.261) | (0.680) | (0.0360) | (0.0145) | (0.0404) |
| Control Mean | 0.503 | 40.851 | 0.036 | 0.038 | 0.226 |
| Optimal bandwidth | 39.25 | 63.53 | 74.81 | 60.30 | 49.76 |
| Observations | 2528 | 2528 | 2528 | 2528 | 2528 |

*Notes:* RD estimates for each (log) outcome with cluster-robust variance at the municipality level (`vce(cluster id)`); both bias-corrected and robust estimates are shown with standard errors in parentheses. A triangular kernel and MSE-optimal bandwidth (`mserd`) are used. The pooled sample covers 2001–2012; the before/after samples are split at 2008. Mato Grosso is excluded. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

**Results and interpretation.** Table G.9 reports RD estimates (cluster-robust at the municipality level). The pre-2008 sample is a placebo: our headline outcome, conflict events, shows no significant pre-2008 discontinuity, and neither do murders. Families affected and attempted murders do show significant pre-period estimates, reflecting well-documented structural differences in frontier land disputes between Pará and its neighbours—differences the SDID absorbs through pre-treatment matching. After 2008, conflict events (−0.129), families affected (−0.945, $p < 0.05$ bias-corrected), and attempted murders (−0.060, $p < 0.01$ bias-corrected) are negative, consistent with lower conflict inside Pará near the border.

The absence of uniformly strong discontinuities is expected for this design rather than as evidence against the main findings. CAR is assigned at the *state* level, so a sharp discontinuity exactly at the municipal border is only one of several margins through which a state-wide reform could appear, and cross-border spillovers in land markets and enforcement plausibly attenuate any discontinuity, biasing the RD toward zero (Vieira and Dahis, 2024). We therefore read it outcome by outcome rather than claiming uniform directional agreement with the SDID. For conflict events, the negative post-2008 discontinuity with no pre-2008 discontinuity is consistent with the SDID. For families affected, the pre-2008 discontinuity (−1.130) is *more* negative than the post-2008 one (−0.945), which taken at face value does not corroborate the SDID's negative sign; we note only that the confidence intervals overlap. We thus treat the RD as a weak, spatially local triangulation whose main value is that it shows no adverse pre-trend in our primary outcome. Causal conclusions rest on the SDID and the alternative-methods checks in Appendices D, Appendix E, Appendix E.3, and Appendix F.

### *G.1. Border-band synthetic difference-in-differences*

As a more direct way to exploit the border, we re-estimate the SDID after restricting the donor pool to control municipalities within a fixed distance band of the Pará boundary, using the signed-distance running variable in Section G. We keep all 144 treated Pará municipalities and run the *same* SDID specification (projected log covariates; same unit and time weights) on the restricted sample. This addresses the concern that the Amazon-wide donor pool may contain poor comparisons for frontier Pará, while retaining the pre-treatment matching that makes SDID more credible than a raw border comparison. Because the restricted pools are smaller than the treated group, placebo inference is unavailable for the tighter bands, so we report bootstrap standard errors (1,000 replications) for all border-band columns.

Table G.10 reports bands of 200, 150, 100, and 50 km alongside the full-sample estimates. The border-band estimates closely track—and for several outcomes strengthen—the full-sample results. For the primary 100 km band (142 donor municipalities), conflict events fall 11.4%, families affected 40.7%, and death threats 10.4%—essentially identical to or larger than the full-sample effects of 11.5%, 33.4%, and 10.0%. Estimates are stable across 200–100 km and attenuate only at 50 km, where the donor pool shrinks to 87

**Table G.10**
Border-band SDID: control group restricted to municipalities within a distance band of the Pará boundary.

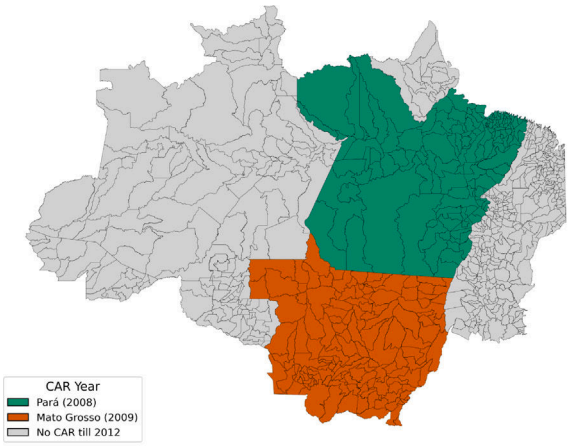
| | Full Sample | ≤ 200 km | ≤ 150 km | ≤ 100 km (primary) | ≤ 50 km |
|---|---|---|---|---|---|
| Outcome: Conflict events | | | | | |
| ATT | −0.122*** | −0.127*** | −0.130*** | −0.121*** | −0.075* |
| SE | (0.030) | (0.035) | (0.036) | (0.038) | (0.043) |
| Relative effect (%) | −11.5 | −11.9 | −12.2 | −11.4 | −7.2 |
| Outcome: Families affected | | | | | |
| ATT | −0.407*** | −0.535*** | −0.554*** | −0.523*** | −0.386** |
| SE | (0.113) | (0.141) | (0.146) | (0.151) | (0.184) |
| Relative effect (%) | −33.4 | −41.5 | −42.5 | −40.7 | −32.0 |
| Outcome: Attempted murders | | | | | |
| ATT | 0.010* | 0.022 | 0.023 | 0.024 | 0.036* |
| SE | (0.006) | (0.021) | (0.020) | (0.024) | (0.020) |
| Relative effect (%) | +1.0 | +2.2 | +2.3 | +2.4 | +3.7 |
| Outcome: Murders | | | | | |
| ATT | −0.019*** | −0.052** | −0.062** | −0.053* | −0.017 |
| SE | (0.006) | (0.021) | (0.026) | (0.027) | (0.034) |
| Relative effect (%) | −1.9 | −5.1 | −6.1 | −5.2 | −1.7 |
| Outcome: Death threats | | | | | |
| ATT | −0.106*** | −0.115*** | −0.112*** | −0.110*** | −0.107*** |
| SE | (0.021) | (0.034) | (0.035) | (0.032) | (0.035) |
| Relative effect (%) | −10.0 | −10.9 | −10.6 | −10.4 | −10.1 |
| Donor municipalities | 488 | 236 | 190 | 142 | 87 |
| Total municipalities | 632 | 380 | 334 | 286 | 231 |

*Notes:* Each block reports ATTs from the same SDID specification as Table 2 (projected log covariates; municipality and year fixed effects), estimated on progressively restricted donor pools. All 144 treated Pará municipalities are retained; columns keep only controls within the indicated distance band of the Pará boundary. The Full-Sample column reproduces Table 2, Column (1). Outcomes are $\ln(1+y)$; relative effects $(\exp(\hat{\tau})-1)\times 100$. Border-band standard errors are from bootstrap inference (1,000 replications, seed 2025), used because the restricted pools are smaller than the treated group; the Full-Sample column retains placebo-based standard errors. $^{*}p<0.10$, $^{**}p<0.05$, $^{***}p<0.01$.

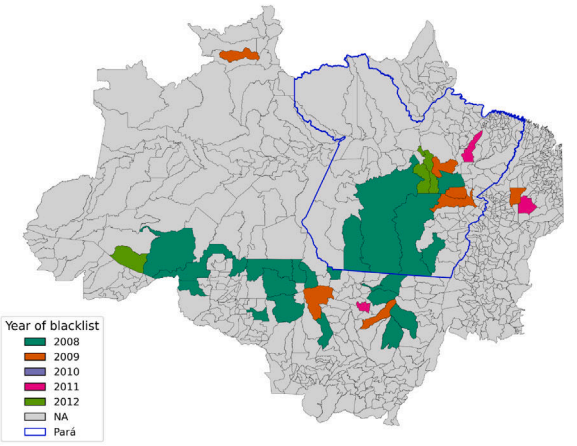
municipalities. Death threats stay near −10% in every band. The rare violent outcomes are noisier, as expected given sparsity near the border. Restricting controls to municipalities directly across the border therefore does not weaken the main findings.

## Appendix H. Key maps

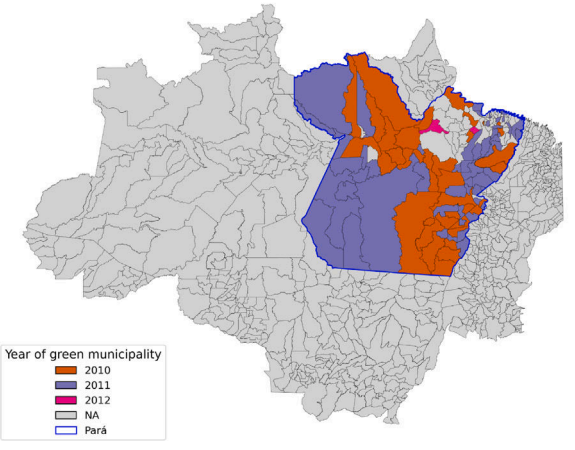
Pará was central to Brazil's early environmental land-governance reforms. Fig. H.9 summarizes the institutional sequence. Panel (a) shows that Pará was the first state to implement the CAR, requiring registration for all rural properties in 2008—four years before the national mandate. Panel (b) shows the rollout of the federal Priority Municipalities program (MP), which targeted high-deforestation municipalities with stricter enforcement and credit restrictions and included one of the largest concentrations of blacklisted municipalities between 2008 and 2012. Panel (c) shows the Green Municipalities Program (PMV), launched in 2010, which expanded gradually across the state with technical and financial support for CAR registration. We exploit this sequence—CAR adoption, MP blacklisting, and PMV—to estimate the effect of CAR registration on rural land conflict, while using MP and PMV to examine the role of complementary enforcement and institutional support.




(a) Early CAR rollout. Pará began in 2008, followed by Mato Grosso in 2009; other states adopted post-2012.




(b) Federal blacklist (MP) rollout, 2008–2012. Municipalities faced increased enforcement.




(c) Green Municipalities (PMV) rollout, 2010–2012. Municipalities received technical and financial support.

**Fig. H.9.** Sequencing of the CAR, MP, and PMV rollouts in Pará. *Source:* Authors' elaboration based on MMA, state of Pará, and IBGE data.

## Appendix I. Heterogeneity and contextual dependence

The estimates reported in the paper represent average treatment effects of mandatory CAR adoption. As discussed in Section 3, however, the mechanisms through which the registry influences land conflict are likely to depend on local institutional and socioeconomic conditions. This appendix clarifies these sources of heterogeneity. They are not separately identified by our empirical design and should therefore be interpreted as contextual factors that bound the external validity of the estimates rather than as additional causal effects.

**Institutional capacity.** The registry generates information, but information changes behavior only when public agencies have the capacity to validate registrations, coordinate across institutions, and enforce environmental and land regulations. This motivation underlies our subsample analyses by PMV participation and pre-treatment enforcement intensity (Table 4), where the estimated effects are larger in municipalities with stronger institutional support.

**Registration timing and registrant composition.** CAR implementation within Pará was gradual rather than simultaneous. Fig. 1 shows that larger properties registered earlier, while smaller landholders entered progressively after 2010. Consequently, municipalities differed not only in the timing of exposure but also in the composition of registrants. Our SDID estimator accommodates staggered changes in municipality-level registration intensity by matching pre-treatment outcome trajectories rather than relying on cross-sectional comparisons (Section 5). Nevertheless, the estimated ATT should be interpreted as an average over this evolving implementation process.

**Property structure and local land markets.** The conflict-reducing effect of registration may also vary with local land structure. In municipalities dominated by large commercial properties, CAR may primarily reduce disputes by clarifying boundaries and increasing the transparency of land claims. In municipalities with many smallholders, registration may additionally facilitate access to formal institutions and public programs. Conversely, where access to registration is uneven, the registry could reinforce existing asymmetries in bargaining power. Our municipality-level design does not distinguish among these channels.

**Historical conflict and local governance.** Municipalities entered the CAR period with different histories of land conflict, civil-society engagement, and state presence. In places where public institutions are viewed as legitimate, registration may facilitate negotiation and dispute resolution. In areas characterized by persistent tenure disputes or limited institutional trust, formalizing claims may initially expose latent conflicts before reducing them. The event-study results are consistent with this possibility, showing a short transition period before conflict declines.

**Implications for interpretation.** Taken together, these considerations suggest that the CAR should not be viewed as having a uniform effect across all frontier settings. Rather, the registry operates through informational and institutional mechanisms whose strength depends on local administrative capacity, enforcement, and land-market conditions. The municipality-level estimates reported in the paper therefore represent average effects under the institutional environment of Pará during the 2008–2012 rollout, an environment characterized by early mandatory implementation, strong links between registration and environmental governance, and complementary support through the Green Municipalities Program.

## Appendix J. Data sources and processing notes

A summary of the data sources, variable construction, and processing procedures used in the analysis is provided in Table J.11.

**Table J.11**
Data Sources and Processing Notes.

| Variable | Source | Notes and Processing |
|---|---|---|
| Land Conflict Outcomes | Comissão Pastoral da Terra (CPT) | PDFs digitized via Python OCR; aggregated to municipality-year counts of conflicts, affected families, attempted murders, murders, and threats. |
| CAR Registration (Pará) | SICAR via secondary sources | Municipality-level indicator for first CAR adoption year (2008); obtained from a prior published study. |
| Municipal Boundaries | IBGE | 2000/2001 shapefiles; reconciled across time in QGIS for consistent geography. |
| Blacklisted Municipalities | MMA/IBAMA | Binary indicator for municipalities blacklisted under federal non-compliance programs (PPCDAm). |
| Green Municipalities (PMV) | PMV Program (Pará) | Binary indicator for municipalities enrolled during CAR rollout. |
| Deforestation & Forest Cover | MapBiomas Collection 6 | Annual $km^2$ from classified land-use data via Google Earth Engine. |
| Environmental Fines | IBAMA | Annual number and value of fines; values in constant 2019 R$. |
| Protected Areas | WDPA and ICMBio | Municipality-level protected-area coverage ($km^2$); federal and state units. |
| Livestock Density | Municipal Livestock Survey (PPM/IBGE) | Cattle per $km^2$, from municipal area and livestock counts. |
| Climate Variables | TerraClimate via GEE | Monthly precipitation (mm) and PDSI, aggregated to annual means. |
| Economic Indicators | IBGE; Assunção et al. (2019) | GDP per capita and price indices; in constant 2019 R$. |

*Note:* Monetary values in constant 2019 R$. Geographic aggregation uses official IBGE municipality boundaries; shapefile and raster processing in QGIS and GEE.

## Appendix K. Construction of covariates for estimation

To maintain a complete, balanced panel, we impute missing values for five time-varying covariates—GDP per capita, mean annual precipitation, mean annual maximum temperature, population density, and cumulative protected area—in a two-step procedure. We first apply within-municipality linear interpolation over time, exploiting the panel structure; we then fill any remaining gaps with the year-specific cross-sectional mean, and, if an entire year is missing (e.g., GDP in 2001), with the overall sample mean. Imputed covariates are then transformed as $\log(1 + x)$ to reduce skewness and admit zeros. Retaining interpolated covariates avoids dropping municipalities, keeps the estimation sample consistent, and follows standard practice when covariates improve precision but are not the object of interest. Table K.12 reports statistics before and after imputation.

**Table K.12**
Descriptive statistics of Key covariates (Original and Imputed).

| Variable | Obs. | Mean | Std. Dev. | Min | Max |
|---|---|---|---|---|---|
| *GDP per capita (R$)* | 6938 | 5776.91 | 5690.18 | 936.16 | 131,150.52 |
| *GDP per capita (imputed)* | 7584 | 5778.15 | 5443.34 | 936.16 | 131,150.52 |
| log(1 + GDP per capita) | 7584 | 8.44 | 0.63 | 6.84 | 11.78 |
| *Mean annual precipitation (mm)* | 7572 | 162.23 | 46.04 | 66.02 | 335.51 |
| *Precipitation (imputed)* | 7584 | 162.23 | 46.00 | 66.02 | 335.51 |
| log(1 + Precipitation) | 7584 | 5.06 | 0.28 | 4.20 | 5.82 |
| *Mean annual max temperature (K)* | 7572 | 316.24 | 7.31 | 285.11 | 341.50 |
| *Max temperature (imputed)* | 7584 | 316.24 | 7.31 | 285.11 | 341.50 |
| log(1 + Max temperature) | 7584 | 5.76 | 0.02 | 5.66 | 5.84 |
| *Cumulative protected area ($km^2$)* | 7584 | 1214.39 | 5951.63 | 0.00 | 92,894.70 |
| *Protected area (imputed)* | 7584 | 1214.39 | 5951.63 | 0.00 | 92,894.70 |
| log(1 + Protected area) | 7584 | 1.69 | 3.05 | 0.00 | 11.44 |
| *Population density (per $km^2$)* | 7569 | 26.21 | 133.99 | 0.08 | 2738.56 |
| *Population density (imputed)* | 7584 | 26.21 | 133.85 | 0.08 | 2738.56 |
| log(1 + Population density) | 7584 | 2.17 | 1.20 | 0.07 | 7.92 |

*Note:* Original variables are preserved; imputed counterparts use within-municipality linear interpolation, then year-specific (or overall) means for residual gaps. Log transforms use $\log(1 + x)$. Sample: 632 municipalities, 2001–2012 ($N = 7{,}584$).

## Data availability

Data will be made available on request.

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